PRAYCG

Master Theory and Protocol

Memory, meaning, methods and evidence

Public research edition v3
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The narrative experiments and their theoretical foundations

Restores the complete PRAYCG3 research account alongside PRAYCG4, the reconstructed theory, corrected measurements and recorded evidence. Includes a section-by-section record of what was retained, corrected or moved.

Contents

1. Purpose, questions and evidence . . . 3
2. Topo-OSM: memory, meaning and maintained organization . . . 6
3. PRAYCG3: from baseline to after-state . . . 13
4. Controls, extensions and stimulus preparation . . . 16
5. Acquisition, hardware and physical timing . . . 19
6. Analysis and signal review . . . 22
7. Recorded studies and reconstruction results . . . 28
8. Decoder Gate: design, predictions and results . . . 31
9. OptoPING: physical feasibility and discriminating tests . . . 35
10. Research workflow, reporting and exchange . . . 38
11. Research milestones and decision criteria . . . 39
12. What v3 preserves and changes . . . 41
Appendix A. Reconstructed mathematical specification . . . 43
Appendix B. Decoder Gate technical specification . . . 50
Appendix C. Optical model and reproducibility testbeds . . . 52
Appendix D. Earlier reconstruction comparisons and terminology . . . 55
Appendix E. Operational measurement definitions . . . 57
Appendix F. Disposition of the original master . . . 59
Appendix G. References and source guide . . . 63

About this edition

PRAYCG asks whether receiving a story produces a measurable physiological and behavioral trajectory that differs from disrupted audiovisual structure and from the same intact material approached as an analytic task. It also asks what continues after the stimulus: recognition, integration, relief, recovery, or a mixture. The experimental distinctions are central to the project.

This master restores the detailed PRAYCG3 research account from public v1.1 while retaining the reconstructed Topo-OSM theory, corrected mathematics and evidence review of v2.4. PRAYCG4 and the Semantic Meaning Gradient remain distinct extensions. Decoder Gate and OptoPING retain their own predictions, assumptions and unresolved tests. The Workbench supports these studies and other research; its sophistication does not determine whether a particular hypothesis is true.

Chapters 1-4 explain the question, theory and experiments. Chapters 5-7 describe acquisition, analysis and recorded findings. Chapters 8-11 cover the newer investigations, research workflow and next tests. Chapter 12 and Appendix F account for every numbered section of v1.1. The technical appendices retain the equations and distinguish an implemented calculation from a proposed interpretation.

The software reference is the packaged Public Alpha 11.11.0 release. Its source inspections, retrospective equipment checks and documented limits are identified separately from the earlier human studies. This document does not add participant data or rerun the archived scientific analyses. Ongoing pilot revisions are not silently adopted. Earlier documents and recordings remain unchanged.

Evidence labels

Preserved proposal identifies the original research idea. **Mathematical correction** repairs a specific algebraic, dimensional or interpretive statement. **New formalization** identifies a declared modeling choice. **Observed result** refers to a saved recording or test within its stated scope. **Empirical background** identifies published work relevant to a component of the proposal. **Implementation finding** describes what a versioned worker actually calculates. None of these labels makes a proposed construct an established measurement.

1. Purpose, questions and evidence

PRAYCG investigates how an experience changes a person's state, how something previously learned becomes available again, and how an organized trace can persist while the system that carries it changes. Its original theory connects encoding, cue-sensitive retrieval and maintenance under molecular turnover. Its human research asks how narrative reception, recognition, integration and after-state can be distinguished from attention, sensory drive and ordinary recovery.

Topo-OSM is the organizing theory, with separate cellular and human-scale formulations. Decoder Gate concerns access and receiver state; the meaning extensions concern consequential updating; OptoPING proposes possible reciprocal bridges between a structural state and electrical activity. These ideas constrain proposed tests, but they are not established mechanisms. Chapter 2 restores their relationship, and Appendix A states the repaired mathematics.

The open, local-first PRAYCG Workbench is the research infrastructure for those tests. It organizes protocols, stimulus preparation, hardware, synchronized recording, analysis, explanations and bundles. It is also intended to support independent studies beyond PRAYCG's own hypotheses. Comprehension, recognition, emotional impact, later recall and physiological persistence remain separate outcomes; a study must measure their relationships rather than assume one score represents them all.

The question behind the protocol

The original master framed the question directly: when a story is allowed to matter, does the body-brain system follow a different trajectory from recognizable structure being disrupted or from the intact story watched under analytic demand? That question remains the starting point [P0]. Its experimental form requires separate observations of what a participant saw, heard, understood, recognized, reported and continued to carry afterward.

A film changes luminance, motion, cuts, speech, music, expectation and emotional context together. Watching it also changes gaze, posture, breathing and muscle activity. PRAYCG was designed to ask which parts of a response survive comparisons with those alternatives. A striking trace during Target is an observation to explain; a pattern across declared contrasts is stronger evidence for a specified prediction.

The original account distinguished six layers:

Layer	Question to investigate
Sensory tracking	How much of the response follows measured audiovisual features?
Local recognizable structure	Do faces, objects, bodies and intact shots explain the difference?
Narrative reception	What does coherent sequence add, and is comprehension actually demonstrated?
Analytic task stance	How does instructed cue monitoring and calculation change the response?
Integration and after-state	What persists, changes or returns after the material ends?
Receiver availability	How do access, prior knowledge, familiarity and present state alter reception?

These layers are related questions rather than a causal decomposition already established by the controls. They give each comparison a purpose and make it possible to state what remains unresolved.

Availability and the receiver

The participant brings a learned history to the stimulus. A cue can evoke a rich response without presenting every detail of the memory again; a fully audible story can be recognized without producing a large new update. Relevance, recognition, comprehension, emotional impact, later recall and physiological persistence therefore remain separate outcomes. Their association is part of the research question.

The original availability-under-load expression combined coherence, adaptive capacity and persistence against mismatch and load. Here it remains an organizing proposal, with no universal numerical scale assigned to those terms. The newer Decoder Gate studies provide narrower operational tests. Their results cannot by themselves identify every process named in the broad proposal.

Participant availability and measurement availability require different records. A person may be unable to follow masked dialogue; alternatively, they may understand it while a poorly coupled electrode prevents analysis. Artifact burden can obscure a response without demonstrating a closed biological gate. No latent gate should be inferred merely because a predicted physiological effect was absent. Independently observed access, behavior, task adherence and uncertainty constrain that interpretation.

The experimental sequence is:

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assigned material and task
  -> observed behavior and physiology
  -> source, timing and quality checks
  -> comparisons with competing explanations
  -> interpretation and a reproducible study record
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The Workbench's contribution is continuity across this sequence: protocol, materials, hardware identity, acquisition, analysis inputs and reports should refer to the same study. It records inconsistencies rather than silently guessing a channel, selecting an unrelated output or treating a planned analysis as completed [W1-W4]. This is useful engineering even while particular research measures remain exploratory.

What the evidence establishes

Evidence	What it establishes	Remaining question
Inspected implementation	A specified workflow or calculation exists	Does it behave correctly on other inputs?
Synthetic or fixture test	Behavior on declared cases, including failures	Does that behavior generalize to real recordings?
Measured functional test	Observed operation of a particular device/configuration	Are calibration, reliability and physical timing adequate for the intended endpoint?
Exploratory human recording	A pattern in the recorded conditions	Do order, stimulus, task or artifact explain it?
Prospective held-out evaluation	Performance of a frozen prediction in a defined design	Does it generalize to new people, materials or settings?
Independent reproduction	Another operator can obtain or reanalyze the result	Could both studies share a measurement or design error?

An unavailable endpoint is not zero; a failed scientific criterion is different from a software crash. Repeated analyses of one acquisition are not replications. Measures sharing inputs are not independent confirmations.

The current evidence does not establish a consciousness measure, semantic-specific neural gate, optical memory mechanism, thermodynamic weight of a thought or clinical effect. It supports inspectable software, specific exploratory measurements, recorded pilot observations and falsifiable proposals. Chapters 2 and 7-9 distinguish the proposals, findings and tests still required.

PRAYCG's candidate contributions include its enforced study identity, measured-reference gamma comparisons, optical timing, affordable multimodal adaptations and particular combinations of experimental controls. Their value should be tested through error prevention, incremental prediction, independent builds and reproducible measurements. Existing signal-processing, provenance and narrative methods are upstream foundations. Neither integration nor a new name alone establishes scientific priority.

2. Topo-OSM: memory, meaning and maintained organization

2.1 The theory this project is trying to test

The original question is how an experience can leave an organized trace, how a later cue can reactivate it, and how that organization can persist while the physical system continues to change. PRAYCG brings this question to narrative experience: what happens when a person recognizes, receives, revises or continues to carry a story? Topo-OSM supplies a proposed account of maintained organization and state change. The recording and analysis tools supply ways to challenge parts of that account. The organizing hypothesis is that experience can alter a distributed, retrievable organization whose maintenance and later effects depend on the receiver's prior state and continuing dynamics.

The theory contains two related but distinct hypotheses. **OSM-CELL** proposes a contribution from intracellular structural states to encoding and later neural response. **OSM-TOP** proposes that human-scale activity can reveal changes in state organization, persistence and reactivation. Neither is established by naming an EEG feature after it. A finding at one level does not automatically identify a mechanism at the other [T1, T5].

This edition restores those questions without treating their original equations as settled laws. It preserves the encoding-storage-retrieval sequence, distributed memory, refresh under turnover, receiver history and meaning-related state change. It also identifies where earlier formulations were revised and where mathematical definitions still require operational measurement.

2.2 Encoding and cue-sensitive retrieval

The early OSM proposal begins with neural activity, calcium entry and biochemical change. A calcium/CaMKII-linked phosphorylation state is hypothesized to alter a tryptophan-related or microtubule response spectrum. These proposed storage sites are part of the original hypothesis, not established memory substrates. A later drive near that altered response could then produce a stronger response than a detuned drive. The intended memory claim concerns a history-dependent change in responsiveness, not simply activity continuing after a stimulus [T1].

The network patch makes encoding conditional on more than sensory intensity. Incoming drive, inhibition, prediction error and plasticity-related gain may jointly influence whether an episode changes the storage variables. Familiar input could be processed or recognized without producing the same update as an unexpected but comparable input. These are model hypotheses: novelty, surprise and neuromodulatory gain are not interchangeable, and increasing voltage does not universally increase calcium current because receptor opening and driving force both matter [T2].

A scalar storage variable can represent a degree of change, but cannot describe an arbitrary memory's content. The distributed extension therefore proposes a vector or field across sites or compartments. Retrieval depends on overlap between the cue and that learned pattern, with competing patterns considered. This preserves an important distinction: a partial cue can elicit a rich response without carrying all the retrieved information itself.

The oscillator example illustrates conditional selectivity. It does not prove that a biological memory is a tuned oscillator, that its response is detectable at the scalp, or that the largest gamma response is successful recall. The source, coupling, competing responses and behavioral retrieval criterion all need independent evidence.

2.3 What persists when the parts are replaced?

The refresh theory addresses molecular turnover directly. A decades-long memory need not contain decades-old proteins; the proposed persistent object is an organized pattern maintained through replacement and reactivation. Its schematic state is [T3, T12]:

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X_mem(t) = [P_tr(t), B(t), W(t), Z(t), R(t)]
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Variable	Proposed role	What must be specified
P_tr	Structural trace or pattern	Observable identity, location and history-dependent change
B	Stabilizing scaffold state	Which measured processes alter turnover or support
W	Synaptic/circuit organization	The task-relevant pattern and its relation to retrieval
Z	Transcriptional/regulatory support	Rebuilding conditions, not a literal memory file
R	Replay, recall or reactivation	Independently identified activity and its maintenance effect

The notation X_mem distinguishes the proposed memory architecture from a score of meaning; P_tr distinguishes a structural trace from the phosphorylation fraction used in Appendix A.1. These are notation clarifications, not additional measured states.

Memory maintenance through reactivation and biochemical feedback already has explicit synaptic models; Smolen's model is one relevant comparison [R8]. The distinct question here is whether the proposed structural variables add independently measurable explanatory value. This is an architecture for a hypothesis, not evidence that every memory requires all five components. Trace strength, information content and retrieval success also remain different quantities. A stable scalar amplitude could conceal replacement by a different pattern; preservation therefore requires a declared fidelity measure as well as persistence.

The earlier shorthand said refresh must exceed loss. That captures a useful intuition but is not a universal threshold theorem for the proposed coupled equations. Continuing input can sustain a nonzero trace even below a nominal refresh/loss ratio, while an unstable feedback loop can amplify activity without preserving memory. The corrected task is to examine bounded dynamics, stable states, perturbation recovery and pattern fidelity under the actual assumptions. Appendix A.3 gives the precise threshold for one saturating self-renewal model and a separate error-correction model. A candidate refresh process earns support through measurements of the proposed variables and their contribution to retrieval.

Empirical background. In mouse hippocampal engram ensembles, Marco and colleagues observed chromatin changes across encoding, consolidation and reactivation [R12]. Das and colleagues observed repeated Arc transcription and local translation after a single stimulation in a neuronal preparation [R13]. These findings make regulatory support and sustained molecular renewal reasonable processes to investigate. They do not identify the proposed OSM structural trace, establish decades of information preservation, or show that every memory needs the five listed components. Turnover rates must be measured for the actual components rather than assumed to be uniform.

Hypothesis still requiring a discriminating test. A specified structural or regulatory variable should improve prediction of trace-specific fidelity or retrieval beyond a credible synaptic and reactivation account that already allows plasticity, turnover and maintenance. Perturbations must verify target engagement, cover the

relevant timescale and control for impaired viability, sensation or retrieval. Failure to recall can reflect impaired access; it is not sufficient evidence that the stored pattern was erased.

2.4 Candidate bridges to neural activity

The membrane-coupling proposal leaves several physical possibilities open: optical, electrical, dielectric or vibrational/structural modes could, if measurable, influence membrane threshold, gain or timing. It introduces a transfer process between an intracellular state and an E/I network, rather than replacing synaptic dynamics [T4].

These alternatives have different tests. An optical pathway needs a source spectrum, photon supply, receptor or transducer, losses and a downstream effect. An electrical or mechanical pathway needs its own source amplitude, transfer characteristics and intervention controls. Success for one route cannot be assumed for the others.

Optical carrier frequency must remain separate from EEG gamma. A hypothetical slow envelope or beat still needs effective electrical transduction and a source-to-sensor path. The absence of such a demonstrated path cannot be repaired by matching two plotted rhythms. The later photon-budget study tests one candidate bridge; it does not replace the broader structural-memory question.

2.5 The explicit human-scale reformulation

Topo-OSM deliberately changed the human-facing terminology. Here **Opto** refers to structured audiovisual input; **Structural** refers to proposed network-state organization or geometry; **Memory** refers to persistence, retrievability and reactivation. This is a reformulation with a different measurement scope, not evidence that the cellular mechanism has been observed [T5].

An inferred state can summarize EEG, autonomic, stimulus and artifact features. Calling that state topological does not itself establish topology. A study must declare whether it measures connectivity, state-space geometry, recurrence or another property, and how that property is distinguished from ordinary amplitude and autocorrelation. Otherwise the result should remain a feature-state proxy.

Likewise, reciprocal prediction between a work signal and a state proxy does not establish a self-maintaining causal loop. Shared feature inputs, common stimulation and measurement noise can create association. Persistence, reactivation and retrieval need their own outcomes; they cannot all be inferred from a coupling score.

Formulation	Preserved contribution	Current status or explicit change
Early OSM [T1]	Biochemical tuning and cue-sensitive retrieval	Conditional mechanism; scalp EEG is indirect
Network patch [T2]	Encoding depends on network state; memory can be distributed	Candidate model; correct the unconditional voltage/current inference
Refresh architecture [T3]	Pattern maintenance across turnover	Retained; replace the universal ratio threshold with model-specific analysis
Membrane bridge [T4]	Candidate intracellular-to-network coupling	Optical, electrical and structural routes remain distinct
Topo-OSM [T5]	Human state organization, persistence and reactivation	Explicitly separated from cellular claims

Formulation	Preserved contribution	Current status or explicit change
Meaning extensions [T6-T8]	Update geometry, adaptive use and stimulus gradients	Operational proposals needing measurement validation
Wild Hare synthesis [T9]	Receiver history connects recognition, update and continuing readiness	Speculative synthesis, not a result or additional physical law

2.6 Meaning depends on the receiver

The meaning-state formulation asks how an experience relates to a person's model of self, others, world or situation. The key-and-decoder metaphor expresses the preserved proposal: the stimulus supplies structured input, while learned history and present conditions shape what it can evoke. An experience may reactivate a familiar interpretation, alter an expectation, resolve an uncertainty or change what actions seem possible [T6, T9].

This account does not require every meaningful experience to cause a large belief update. A familiar scene can still matter deeply when its content is already expected. Recognition and relevance therefore remain measurable dimensions in their own right. A probability divergence is one candidate measure of change within this broader account. It does not replace the account with a single universal unit of meaning.

Dimension	Candidate observation	Interpretation boundary
Recognition and reactivation	Independently scored recognition and cue-specific retrieval	Familiarity or a returning EEG pattern alone does not establish retained content
Personal relevance	Declared ratings and independently assessed choices or consequences	Relevance does not establish accuracy, benefit or a shared response
Belief updating	Before-and-after probabilities about the same specified outcomes	Large change can be mistaken, harmful or unrelated to the intended construct
Retention and transfer	Delayed recall or performance on new items	Immediate comprehension does not establish durable integration
Interpersonal understanding	Inferences compared with another person's independent reports	Similar responses and physiological synchrony are insufficient on their own

These observations are reported separately. Combining them would require a new, justified measurement model, with weights and missing-data rules fixed before evaluation. Their relationship is a hypothesis to test.

Mathematical correction. D198's posterior-to-prior Kullback-Leibler divergence is retained as model-specific updating, U_i, rather than being called meaning itself. Its relationship to Bayesian surprise has established precedent [R9]. Probabilities must describe the same state space and be collected or estimated by a declared procedure. An arbitrary vector of impact ratings cannot be inserted into the divergence as if it were a probability distribution. Appendix A.7 specifies support, units and estimation limits.

Preserved interpersonal proposal with a narrower name. D198's metric cosine becomes update alignment. It measures whether two reliable change vectors point in similar directions in a shared, fixed coordinate system. Two people accepting the same false claim can align strongly. A person who understands another's distress may respond with calm concern rather than the same emotional change. Empathic accuracy therefore requires a separate comparison between an observer's inference and the target's own independently collected report, with report error acknowledged [R10]. Concern, affect sharing and helping behavior also

need their own observations. The alignment score is not an empathy scale until that interpretation survives such tests.

Predictive validity remains separate. Before-and-after forecasts should be scored against the same later outcomes that were unavailable when the forecasts were made. An improvement can support a limited prediction claim. Feeling certain or internally coherent does not establish that improvement. Recall of a fictional story measures accuracy about that story, not truth about the external world. High reported significance alongside a false belief would be compatible with the proposed separation; one such result would not validate the whole theory.

Communication in these studies uses ordinary sensory and social channels. Relational geometry names a declared pattern of relationships or state coordinates. It does not establish quantum entanglement, a universal entropy law favoring social connection, or a physical quantity of meaning in rocks or stars. Those philosophical extensions do not follow from D198 or from the PRAYCG measurements. The human research remains centered on individual reception, recognition, update and after-state; cross-person alignment is an optional additional question.

2.7 Recognition, integration, peaks and resolution

Recognition and durable update need not coincide. A familiar scene might activate an existing pattern without adding much new information. Conversely, understanding something at first exposure need not produce lasting retention. The research question is whether separately measured comprehension, later retrieval, prediction or transfer distinguish these possibilities. A missing theta endpoint cannot establish that no learning occurred.

The peak/resolution distinction preserves another part of the original intent. One response might involve abrupt accommodation to an unexpected interpretation; another might involve the resolution of a previously active uncertainty. Their immediate intensity, regulatory demands and later trajectories could differ. These remain competing descriptions to test, not types already established by the historical recordings [T9].

For either account, persistence must be distinguished from fatigue, ordinary recovery, continued attention, rehearsal and artifact. Re-presenting an appropriate cue could test reactivation; independently scored retention or transfer could test consequences. Washout is useful observation time, but a slow physiological return alone does not identify what was retained.

2.8 Meaning gradients and constructive updating

The semantic-gradient proposal separates coherent low-human-stakes material from chaotic randomness and from the phase-scrambled sensory control. It asks whether independently assessed stimulus properties and participant responses predict different trajectories. The proposed density and carryover indices remain declared measurements, not objective units of meaning. Stimulus coding should be independent of the physiological endpoint. Manipulation checks must not discard a run merely because its predicted effect failed; that would select evidence using the outcome being tested. Low meaning, high meaning and effective arithmetic override require evidence rather than labels [T8].

Constructive meaning adds a different question: does an interpretation help someone generate usable options, respond to feedback and revise a failing policy? Feeling better, producing more options and making better predictions are separable. D203's intended test includes costs, validity and flexibility, rather than defining every emotionally positive response as adaptive [T7]. The appendix offers one explicitly versioned alternative denominator while preserving signed benefits and costs separately.

2.9 Predictions at the appropriate scale

Hypothesis	Discriminating observation	What would weaken the specified account
Cellular tuning affects retrieval-like response	Measured state change predicts selective response to matched versus detuned drive	No reproducible tuning or specificity after preparation controls
A structural mode affects excitability	Independent perturbation changes the proposed mode and membrane response	Effects explained by heating, toxicity, common drive or measurement pickup
Refresh preserves a trace through turnover	Pattern fidelity and retrieval track specified maintenance processes	Prespecified disruptions leave the predicted contribution unsupported
Human state organization carries information beyond current input	Frozen state features improve held-out prediction of reactivation or behavior	Simple stimulus/history models perform equally well
Recognition and new integration can dissociate	Separate recognition and later-update outcomes vary as predicted	The proposed distinction adds no reliable predictive value
Meaning-frame benefits depend on updating	Improved usable options and feedback adaptation beyond affect/task alternatives	Positivity increases without the predicted behavioral benefit

Cellular tests require appropriate preparations and direct measurements. Human narrative studies can test behavioral and physiological predictions without assuming a molecular bridge. Neither scale should borrow the other's evidence to fill a missing step. The restored theory is a connected research program about maintained organization and consequential experience, with each bridge accountable to its own test.

Additional rejection criteria for this reconciliation. A study must choose which claim it is testing. A null result for an EEG proxy does not refute every possible account of meaning or memory. Conversely, a broad theory cannot be protected by switching its endpoint after a failed prediction.

- An update-only definition of meaning is inadequate if independently reliable relevance or recognition remains high while the specified probabilities do not change. This challenges that definition, not the possibility of familiar meaningful experience.
- An empathy interpretation of update alignment is weakened if it adds no held-out prediction of independently measured empathic accuracy beyond shared stimulus, response style and task. Wide uncertainty remains inconclusive.
- A truth-tracking prediction is weakened if posterior forecasts fail to improve the prespecified proper score on the same held-out outcomes, despite adequate measurement precision. The size of the update cannot substitute for this test.
- A specified structural refresh model is weakened if its measured variables add no predictive value over credible maintenance alternatives, or if a selective, adequately long perturbation leaves its predicted fidelity effect absent. Unmeasured variables or an ineffective intervention leave that test unresolved.

2.10 Revision register

Earlier expression or treatment	Revision in v2.3	Consequence for the claim
Theory largely absent from the recent master	Restore encoding, retrieval, turnover, receiver history and meaning-state change	The original questions again organize the research account
Human network-state formulation treated as sufficient coverage of OSM	Separate OSM-TOP from OSM-CELL	A human feature-state result cannot establish a cellular substrate
OptoPING described only as a photon-receptor loop	Restore the broader structural/membrane bridge; retain photon accounting as one candidate	A photon-budget failure constrains that specified pathway, not every structural hypothesis
Voltage increase assumed to imply calcium-current increase	Include both unblock and driving force	The inequality holds only under the stated derivative condition
Response peak placed at the natural frequency	Give the exact damped peak and the weak-damping approximation	Retrieval selectivity depends on damping and support
Refresh/loss ratio treated as a general persistence threshold	Give bounded dynamics and a model-specific threshold, with a counterexample	Driven persistence, autonomous stability and memory fidelity are separate
Noisy phase locking treated as a simple threshold	Separate noiseless locking from noisy finite-window concentration	Phase slips and common input must be considered
Unsigned coupling magnitude interpreted without dynamics	Retain sign, units, state scaling and stability conditions	Recovery does not establish a stable or causal biological loop
Meaning arc adds variance to an unscaled derivative	Introduce a fixed timescale or report components separately	Dimensional repair still requires validation of the endpoint
Meaning-efficiency denominator can cross zero	Use a separately versioned positive-cost candidate and retain signed components	This is a new operational choice, not a retroactive rescore
Early Rung 0 executables not covered in the recent account	Include recovered initial and C-I models	Trajectory, feature and score tests retain their different evidential scope

The v2.3 reconstruction remains preserved. The following additional revisions define v2.4.

Source claim or shorthand	Revision in this edition	What remains hypothetical
Distributional displacement called meaning	Name it belief updating and retain relevance, recognition and reactivation separately	Which observations best explain meaningful experience
Interpersonal cosine called empathy	Name it update alignment and require an independent empathy criterion	Whether alignment predicts empathic accuracy or concern
Truth tracking expressed using unspecified expected error	Score fixed forecasts on the same held-out external outcomes	Whether the proposed update improves predictive validity
Refresh greater than turnover treated as a general threshold	Retain the explicitly bounded, model-specific threshold and external-support counterexample	Which maintenance processes operate in a biological trace

Source claim or shorthand	Revision in this edition	What remains hypothetical
Copy error plus noise divided by correction gain	Supply a separate stochastic model with distinct bias and variance terms	Its error coordinate, rates, noise model and biological adequacy
Similarity or stable strength treated as retained identity	Require a declared content-fidelity and retrieval criterion over a stated duration	Whether an identified pattern carries the proposed information

All new mathematical choices are candidate definitions. None was fitted to a participant in this revision. No historical module definition or recorded result is silently replaced by these candidates. To use a new equation in a study, first publish its version, measurement procedure, units, parameter choices, missing-data rules, simpler alternatives and rejection criterion. Implementation and prospective validation would be later work.

3. PRAYCG3: from baseline to after-state

3.1 The experiment's central question

PRAYCG3 asks whether an intact story watched naturally produces a different trajectory from a version with recognizable narrative structure damaged, and from the same intact material watched while performing an analytic task. The object of study is the trajectory: access to the material, recognition during viewing, subsequent processing, and the state carried into the periods after viewing. No single increase in EEG activity establishes that this sequence occurred [P0].

The original three-branch design remains the clearest introduction to that question. PhaseScrambled Control challenges a sensory-drive explanation. Target presents the intact story under natural viewing instructions. Contextual Override preserves that material while changing what the participant is asked to do. Baselines, washouts and reports make the transitions observable. Their roles differ, and their observations must remain identifiable in the study record.

3.2 The complete sequence

The legacy PRAYCG3 order is fixed: Control, Target, Override. The following sequence restores its full structure, including a consequential detail of the current implementation: the Override sum response precedes its washout, while the branch experience report follows the washout [P0, W5].

Stage	Participant activity	Research purpose
Preparation	Confirm display/audio access, instructions, recording readiness and assigned material	Establish what can be presented and measured
Baseline 1	Remain quietly at fixation	Describe the initial state and provide eligible calibration support
PhaseScrambled Control	Watch naturally	Observe response to transformed sensory material
Washout 1, then report	Remain quietly, then describe the experience	Separate post-Control physiology from report activity
Target	Watch the intact cued story naturally; ignore the digits	Observe reception under natural viewing instructions
Washout 2, then report	Remain quietly, then describe the experience	Observe the immediate post-Target trajectory

Stage	Participant activity	Research purpose
Contextual Override	Watch the same cued story while adding the digits	Observe the effect of assigned analytic work
Sum response and stillness	Enter the final sum, then settle	Record task performance and isolate response activity
Washout 3, then report	Remain quietly, then describe the experience and task	Observe the post-task trajectory with its intervening response documented
Baseline 2 / final reflection	Quietly reflect on and compare all completed branches	Observe the cumulative state under a declared reflection instruction
Final report	Give comparative ratings and relevant context	Preserve the participant's retrospective account

Instructions and a stillness reset precede each video; a further reset precedes final reflection. The inspected runner defaults to 120 seconds for Baseline 1, each washout and final reflection, with an 8-second stillness reset. These are configurable software defaults, not established optimal durations. The locked study configuration and actual events determine what occurred. Final reflection can be disabled; its absence must remain an absence, not a reconstructed measurement [W5].

An initial baseline is a reference observation, not a state without thoughts, expectations or autonomic variation. Baseline quality, duration and usable coverage determine whether a subsequent normalization is supported. The recording must retain instructions, video boundaries, responses, washouts, reports and deviations as distinct events. Planned timing alone does not establish physical presentation timing.

3.3 What the three branches can distinguish

PhaseScrambled Control is a meaning-damaged sensory control. It may preserve selected low-level properties while disrupting recognizable audiovisual organization. It can still provoke entrainment, scanning, boredom, discomfort, confusion, affect or attempts to discover a pattern. Its physiological activity is part of the comparison. Calling it meaningless would mistake an intended manipulation for an observed participant state [P0].

Target is the intact stimulus watched naturally. The instruction permits reception without requiring an emotional response. A participant may follow the story, recognize familiar content, disengage, or experience little personal relevance. These possibilities belong in the data. Target activity becomes informative through comparisons, behavioral evidence and examination of plausible sensory and artifact explanations.

For the instruction-only Target/Override contrast, both branches use the same cue-embedded audiovisual file. The number overlays, their encoded positions and timing therefore match; Target instructs the participant to ignore them, while Override requests a running sum. The current manifest declares Target and Override a bit-identical media group. Different filenames do not establish equivalence, and an uncued Target compared with a cued Override would change the intended contrast [P0, W5].

Override tests task stance. It changes cue monitoring, working memory, arithmetic, effort, visual attention and possibly gaze or muscular tension. It may reduce narrative absorption, leave comprehension intact, or help a participant notice material missed earlier. Meaning may also break through the task. A Target/Override difference consequently supports an effect of the assigned task under the recorded conditions; identifying its psychological pathway requires additional measurements. Correct arithmetic alone does not prove that the story was excluded from processing.

3.4 Washouts and the final reflection state

Washout preserves observation time after a branch. It does not guarantee erasure of the preceding experience or restoration of Baseline 1. A story can remain active through imagery, rehearsal, emotion or reflection. Breathing can settle, task effort can subside, and physiology can recover on a different timescale from subjective experience. Those alternatives motivate measurement of both the immediate trajectory and its supporting context [P0].

Reports follow the washout so ordinary rating and typing activity is separated from that observation interval. Override has an additional asymmetry: sum entry and a stillness reset occur between video offset and washout. Post-branch comparisons must preserve elapsed time from video offset and account for that intervening task. Equal washout labels or durations do not make their beginnings equivalent. Response and report periods should not be silently included in passive after-state endpoints.

Baseline 2 is particularly informative and particularly easy to overinterpret. The current instruction asks for quiet reflection on all three blocks. It is therefore a cumulative reflection condition, not an untouched resting baseline or a Target-only follow-up. It can contain narrative carryover, task-completion relief, fatigue, habituation, respiratory change, deliberate comparison and memory reactivation. Its difference from Baseline 1 also includes a difference in instruction [P0, W5].

Attribution should examine earlier washouts, task performance, participant reports and autonomic trajectories together. Resemblance to post-Target activity can motivate a hypothesis, but does not identify its cause. A strong delayed effect with a weak in-clip peak can be compatible with later processing or ordinary recovery; the distinction needs an outcome and comparison capable of separating them.

3.5 First-person reports as a separate evidence stream

Self-report preserves what the experience was like for the participant. It can agree with physiology, diverge from it, reveal sensory problems, or indicate a strategy the experimenter did not anticipate. Reports are separate measurements, not automatically statistically independent evidence: the same stimulus, participant history, task and expectations can affect both report and physiology [P0].

The six core branch domains are personal meaning, absorption, emotional afterglow, story activity during washout, task extraction load and confound burden. Supporting context includes audio comprehension, audiovisual synchrony, cue legibility, eye strain or squinting, external noise, familiarity and relevant prior exposure. Override additionally needs task performance and the participant's account of effort and story breakthrough. Final comparisons preserve which branches were experienced as more meaningful, demanding or persistent. The study should distinguish a question that was not administered from a low response [P0, W5].

Scene descriptions should identify a line, image, sound or felt moment without demanding an estimated clock time. Retrospective timing guesses cannot supply precise prospective anchors. Questions should also allow no meaningful moment, no afterglow or an unsuccessful manipulation. Excluding participants because their meaning ratings failed to follow the predicted ordering would select evidence using the outcome the study seeks to explain.

Reports remain fallible, influenced by recall, wording and the demand to explain. Their value is increased by preserving prompts, response timing and missingness, and by keeping comprehension, personal relevance, arousal and task effort distinguishable. A disagreement between report and physiology is a result to investigate; neither stream automatically overrules the other.

3.6 Recognition, integration, peaks and resolution

The original account distinguished recognition from integration and an acute peak from a later resolution trajectory. These are useful research axes, not validated participant types. Recognition concerns identifying or reactivating something already available. Integration concerns a proposed change in organization or consequence that must be specified through outcomes such as later retention, retrieval, prediction or transfer. Familiarity can strengthen recognition while reducing novelty; it does not make an experience personally insignificant [P0].

The peak/resolution axis concerns temporal form. An abrupt recognition or reinterpretation may produce a brief candidate event. Closure, repair or resolution may develop more gradually or continue after the clip. Immediate intensity, subsequent calm and retained information need not covary. A slow physiological return does not reveal what was learned, and an unavailable theta endpoint does not establish that learning failed.

The pilot-derived descriptions remain hypotheses: recognition without a clear after-state, an acute event followed by later density, modest in-clip impact with stronger subsequent activity, or a delayed response despite compromised sensory access. The last possibility motivates a decoder-availability question: did partial input reactivate prior knowledge? Each pattern also admits sensory, task, respiratory and artifact alternatives [P0].

CAI/SID can provide a secondary description of declared access-like and integration-like features over supported time [M1]. A peak score, an interval density and a participant report answer different questions. Shared inputs prevent their agreement from counting as independent replications. The inference becomes stronger when a frozen physiological prediction explains a separately measured consequence on held-out data, beyond credible stimulus and participant-history comparisons.

4. Controls, extensions and stimulus preparation

4.1 PRAYCG4 and the structural-control question

PRAYCG4 extends the three-branch sequence by adding ShotOrderScramble, with its own washout and report. In the legacy order it follows PhaseScrambled Control and precedes Target. The purpose is specific: phase scrambling damages local recognizable structure as well as narrative continuity. A Target/PhaseScrambled difference can therefore reflect faces, bodies, objects or biological motion without requiring a narrative-order explanation [P0].

ShotOrderScramble retains intact local shots while changing their larger order. Depending on the declared audio treatment, it may also preserve local speech fragments. It supplies a middle comparison with more recognizable content than phase scrambling. Reordering nevertheless changes continuity, transitions, expectancy, confusion and sometimes cut rhythm. Participants may reconstruct a familiar plot or find a new interpretation. The manipulation does not isolate meaning perfectly.

A prospective hypothesis may predict a specified Target endpoint exceeding PhaseScrambled, ShotOrderScrambled and Override at locked windows. That direction must belong to the endpoint and theory being tested; every physiological measure need not increase with reception. Report each contrast, uncertainty and nuisance sensitivity. The pattern across controls is more informative than choosing whichever comparison happens to favor the theory.

4.2 Semantic Meaning Gradient

The Semantic Meaning Gradient is a separate family. Its original sequence compares coherent low-intended-meaning material, a high-meaning target and arithmetic override. The low-meaning condition should not simply be random noise: randomness can create novelty, threat, puzzle demand or prediction error. Coherence and personal meaning are distinct stimulus and participant properties [P0].

Candidate stimulus coding includes social presence, agency, vulnerability, causal continuity, care or repair, stakes and transformation, with overload and confusion considered separately. These dimensions require declared coding rules and independent characterization. A category name, weighted index or threshold does not create objective units of meaning. Measures of comprehension and participant relevance help establish whether the intended manipulation occurred without defining success by the physiological result.

This family asks whether characterized content and receiver context predict different trajectories. It does not replace the sensory and structural comparisons of PRAYCG3/4. Selecting an accessible story, disrupting its order and changing the participant's task are different experimental interventions.

4.3 Order, repetition and familiarity

The fixed PRAYCG3 sequence preserves a historical feasibility design but confounds branch with position. Target normally precedes the intact story's repetition under Override. Familiarity, prediction, learning, fatigue, boredom and carryover can therefore contribute to their difference. Longer washouts do not necessarily remove those influences [P0].

Prospective sequence variants support counterbalanced assignment. Use the actual declared sequence and allocation rule, not the protocol family name, to document balancing. Williams-style or other balanced orders address specified positional and first-order carryover effects when implemented with suitable allocation; they do not eliminate persistent learning, nonlinear carryover or cumulative fatigue [W2, W5].

Record prior viewing, knowledge of the scene, exposure to related material, repeat sessions and relevant context. Novel material can test first-pass reception; familiar material can test reactivation. The same clip can serve as a new story for one person and a cue to an established memory for another. Changing a participant identifier does not erase that history.

Separate development, calibration and evaluation. Hold out whole participants, sessions or stimulus families according to the intended generalization. Overlapping windows from one recording are not independent replications, and random window splitting can transfer nearly identical information into training and evaluation. A study should declare its unit of inference before interpreting a large number of windows as a large sample.

4.4 Anchors, null windows and prospective claims

Anchors specify where a candidate recognition, integration or resolution event is expected. Primary peak anchors, secondary context anchors, resolution anchors and clip-edge anchors serve different purposes. Define each window and endpoint before acquisition or before an explicitly independent evaluation. Frame-verify times on the final rendered Target, freeze the anchor file and record its identity with the study [P0].

An anchor chosen from the story is a timing hypothesis. An anchor chosen after inspecting the same physiology is exploratory. Participant descriptions can contextualize either, but must not retroactively convert the latter into a prospective test. For an event near the clip edge, specify whether the endpoint can extend into the washout and whether its analysis method permits crossing that boundary. A continuation hypothesis cannot override a method's requirement for support within one continuous block [M1].

Null or pseudo-anchor windows test how often comparable candidates appear elsewhere. Their selection should respect window duration, usable coverage, temporal dependence and relevant sensory structure. An easy null can exaggerate apparent specificity. Anchor counts, endpoint multiplicity and selection rules require a declared statistical treatment. A successful locked-window result still needs controls and held-out evaluation; freezing a file prevents drift but does not validate its interpretation.

4.5 Preparing the stimulus family

Select material with documented source identity, permitted use, content considerations and enough context for the intended audience. The original practical range of roughly 2.5-5 minutes was a stimulus-selection guide, not a validated universal optimum. Clip length should support the intended event, after-state observation and participant burden. Public software distribution does not grant redistribution rights to film excerpts or participant recordings [P0, W2].

Begin from a retained master. Apply any justified cleanup consistently before generating the branches, and document subtitles, source stamps or other overlays. Generate the cue-embedded intact Target and use that same file for Override. Generate the phase-scrambled control from the intended cued source so the transformation includes the cue contribution. Normally generate ShotOrderScramble from the cleaned master and then apply the same cue schedule, retaining the permutation seed and shot table [P0].

The stimulus package should bind the master and derived files to the cue schedule, transformation settings, preparation manifest, visual/audio review, locked anchors and fingerprint outputs. Preserve original materials alongside derivatives. A failed or changed export is a new artifact whose timing and identity must be checked again.

4.6 Quality review and fingerprints

Manual viewing should verify playability, duration, frame cadence, tail preservation, cue legibility, overlay treatment and branch identity. Target and Override must meet the declared identity requirement. ShotOrderScramble needs inspection of its actual reordered material, not only a successfully generated permutation. Review the resulting experience for unusually long discontinuities, unintended freezes or other changes that could dominate the comparison [P0, W2].

Control audio requires human listening. Review for recognizable words, sentence fragments or distinctive melodic material that could identify the source, alongside the intended loudness and envelope relation. Automated processing cannot certify unintelligibility. Record who reviewed what version and any residual cues. A failure of the intended manipulation must remain visible rather than being repaired by relabeling the branch.

Fingerprints describe measured audiovisual properties such as luminance, motion, cuts and audio features. Bind each extraction to the exact file, transformation, extraction version, units, time grid and missingness. Shot reordering requires its own temporal mapping; the intact film's timestamps cannot simply be reused [W2].

Fingerprint adjustment asks how much a physiological association changes after accounting for specified measured stimulus features. It does not remove all sensory explanations. Show unadjusted and adjusted results, assess model adequacy, and acknowledge unmeasured features. A weak nuisance model cannot establish that the nuisance is absent. A file hash establishes identity; it does not establish successful playback, physical timing or semantic equivalence.

4.7 From a prepared protocol to an interpretable study

Lock the protocol identity, version, ordered branches, instructions, configured durations, media, cue schedule and anchors before recording. Preserve runner identity and actual events. Acquisition should connect these records to the EEG, available autonomic and respiration streams, markers and physical timing evidence. The light sensor measures display behavior; it does not measure a biological response. Suitable eye, muscle or motion measurements can test artifact explanations that scalp EEG alone leaves unresolved [P0, W2].

The original research architecture joined preparation, acquisition, analysis and review so they refer to the same experiment. That continuity remains essential. A completed run, a valid timing observation and an available analysis endpoint are different achievements. Report missing support and failed quality criteria before interpreting trajectories.

The next prospective test should choose its question, contrasts, anchors, outcomes and exclusion rules in advance; obtain adequate sensory, timing and artifact evidence; preserve self-report and task performance; and evaluate the frozen claim on new observations. Recognition, task-dependent reception and persistent consequences can then be tested in their own terms. The protocol earns support by distinguishing these possibilities, including when a simpler explanation accounts for a promising pattern [P0].

5. Acquisition, hardware and physical timing

5.1 What the equipment can currently support

PRAYCG brings the protocol, prepared material, equipment and recording evidence into one study. Hardware support still has a specific scope: a working connector establishes a software route; a successful recording demonstrates that route in the observed configuration; a scientifically useful endpoint also needs suitable placement, timing and signal quality. The current hardware evidence does not cover every unit, firmware or installation [H1, H2].

Hardware or route	Current capability and evidence	Remaining boundary
OpenBCI Cyton+Daisy, optional ALS/PT19	Existing recording route: 16 EEG channels at nominal 125 Hz, auxiliary/light-sensor feeds and status markers; catalog bench verification.	Review actual electrode contact, contamination and physical timing.
Polar H10 RR	Measured beat-interval durations through BLE, with connection boundaries preserved.	Arrival timestamps are not measured beat endpoints. The separate ECG route has software/simulator evidence.
Vernier respiration belt	Raw force/movement acquisition; packaged 10 Hz profile and explicitly configured 20 Hz option.	Host-derived timing and placement-dependent amplitude do not establish physical latency or breath volume.
Muse S Athena, frozen BrainFlow p1041 route	Single-unit acquisition/recording evidence; four EEG electrodes at 256 Hz, separate motion and raw optical feeds.	No ALS; transport evidence does not qualify absent electrodes or optical hemoglobin conversion.
Compact Cerelog, 3 EMG + 2 EOG + ALS	Provisional functional evidence and managed experimental acquisition at nominal 1 kHz.	Combined EEG/reference timing and artifact coverage remain to be qualified.
Legacy Cerelog 4 EMG + ALS and 5 EMG + 2 EOG + ALS	Separate experimental profiles and pilot routes.	They do not inherit compact-device evidence or expanded authority.

Hardware or route	Current capability and evidence	Remaining boundary
Crown/BrainFlow OSC and managed GazePoint GP3/GP3-HD	Exact software routes, with software/simulator evidence where documented.	Physical device validation remains outstanding.
Other Muse/OpenMuse, Pupil, Emotiv and existing-LSL publishers	Product-specific external-publisher and stream-contract pathways.	Discovery or catalog presence does not establish complete compatibility.

5.2 The compact Cerelog configuration

The compact route requires `CERELOG_V1_EMG3_EOG2_ALS1_1K_V1` firmware and its declared channel/gain configuration:

- **CH3:** left temporalis EMG; **CH4:** right temporalis EMG; **CH6:** right frontalis EMG.
- **CH7:** designated horizontal EOG; **CH8:** designated vertical EOG. These are placement labels; clean separation of the axes has not been demonstrated.
- **CH5:** ALS light sensor. **CH1 and CH2 are disabled**, not physiological channels.

EMG/EOG use gain 24 and export microvolts; ALS uses gain 1 and exports volts. Conversion assumes a nominal 4.5 V reference rather than an independently measured amplitude calibration. Raw eight-slot counts remain available. This arrangement lacks the two masseter channels in the older five-EMG configuration [H1].

The retained 90-second functional recording described in v2.4 contained **89,752 records**, without detected runtime corruption, serial error flags or active-channel clipping. It contained **59 device-timestamp gap observations**, with a largest forward delta of 2 ms. The format has no sequence counter, so exact missing-sample counts remain unknown.

Repeated bilateral temple bursts were compatible with reported clenches. Earlier recordings showed forehead, eye-task and light-cover/uncover responses. Delayed chat-delivered movement cues prevent precise response-latency or controlled rest-clench-rest claims. Together, these observations support a provisional functional pass and integration testing; they do not establish calibrated muscle responses, electrical safety certification or neural-gamma validity.

The managed connector separates off-body checks from experimental on-body acquisition. The latter requires a reviewed experimental-participant profile and fresh confirmations. Raw serial evidence, gaps and transport events are retained. Local capture complements the multimodal LabRecorder XDF rather than replacing it.

5.3 Timing is a measurement

Media time, runner callbacks, LSL timestamps, transport arrival and screen luminance describe different events. ALS/PT19 records a physical light response in its own acquisition path, providing delivery evidence beyond software callbacks. It does not establish EEG contact or biological meaning.

The passive dual-ALS comparison fits offset/drift using explicitly matched calibration-event IDs, then evaluates separate held-out pairs. Participant, source, acquisition and segment identities, missing events, gaps and uncertainty remain attached to the result. A common clock label establishes neither common participant identity nor physical sensor synchronization. Alignment must not be optimized to maximize a neural contrast.

Gamma Scalpel's separate dual-ALS helper fits reviewed pulse pairs and builds an engineering envelope from residuals and sample brackets. That envelope is neither a statistical confidence interval nor held-out validation. The full participant-connected qualification of the compact Cerelog/Cyton-Daisy arrangement remains outstanding [H1, H2].

Retrospective equipment update from Public Alpha 11.11.0. A corrected detector initialization was applied to the saved six-minute off-body dual-ALS recording. It matched 22 barcode events, with 11 assigned to fitting and 11 to held-out evaluation. The held-out 95th-percentile relative timing error was 7.3006 ms. All nine declared threshold combinations met the original limits; the largest error across that sensitivity review was 18.5398 ms. The historical review and original recording were preserved [H3].

The detector repair and evaluation were retrospective. The held-out event split does not make the overall development process prospective or establish new participant-connected performance. A new full-duration equipment recording and combined body pilot remain necessary. The same release reopens expected-versus-recorded inventories, reporting missing streams, identity/units issues and timestamp anomalies separately from physiological suitability. A bench XDF without the required runner evidence remains an equipment record rather than a completed protocol study [W5, H3].

Device-specific timing limits also matter. Athena's receipt covers connection, EEG identity/units, advancing timestamps, monitor/replay, XDF reopening and disconnection. Its **16 raw optical channels are not 16 anatomical sites**; no verified Athena HbO/HbR conversion profile is bundled. Transport evidence does not establish electrode coupling or physiological accuracy.

Vernier defaults to **100 ms/10 Hz**; **50 ms/20 Hz** requires matching configuration and metadata. The bridge publishes aligned SDK-buffered measurements using approximate host-batch backfilled timestamps. Ambiguous overlapping batches are not silently repaired. Raw force remains distinct from the vendor's derived respiration rate.

Polar RR durations are measured intervals, but intervals arriving together in one BLE notification share its arrival timestamp. Received-order summaries can remain useful while precise event-locked or cardiac-respiratory endpoints require stronger timing evidence. Successive interval differences must preserve original beat adjacency and must not cross reconnect or malformed-packet boundaries [H2].

5.4 Preserve the study and its observations

Before launch, bind the session identity to its protocol, permitted baseline/washout settings, prepared material and selected hardware. Changes after locking need a recorded revision. A shortened demonstration must retain its actual duration and purpose. Bench mode supports software demonstrations; exploratory Athena recording remains a legitimate acquisition mode, even when a full-cap endpoint is unavailable.

Keep the original XDF, events, channel/source map, anchors, media manifest and available transport evidence. Record cancellations and incomplete acquisitions honestly. If LabRecorder continues after the protocol ends, preserve the original and select a reviewed analysis interval.

Choose prospective anchors independently of the physiological candidate. Record who chose each anchor, the material/version, frame/time and whether selection preceded acquisition. Pseudo/random anchors can test how often a candidate occurs elsewhere. HED descriptions preserve original events; syntax validation does not establish physical occurrence or timing. Response-bearing files require privacy review [W2].

Behavioral observations need equally clear identities. Preserve question wording, timing, familiarity, effort, absorption and other ratings separately. Comprehension, recognition and delayed recall assess different outcomes; immediate testing can influence later recall. A rating reused inside a composite cannot independently validate it, and an unavailable EEG endpoint does not erase valid behavioral evidence.

5.5 The shielded enclosure

Eyes-open/closed material already exists from an enclosure described by the operator as shielded, but its attenuation was not independently characterized in the inspected contract. The planned basement/enclosure PRAYCG4 comparison should measure interference and other artifacts under documented conditions. Keep calibration and tuning separate from the recordings used to judge preservation. The enclosure is a measurement opportunity, not clean ground truth.

6. Analysis and signal review

6.1 From a recording to an analysis plan

Analysis Forge begins with the completed recording and its associated evidence. Review the run, resolve ambiguous XDFs or sidecars, run required integrity, event/timing, EEG-quality and line-noise checks, then build the supported plan. Behavioral-only Decoder Gate studies use their dedicated report route without an invented XDF [W1-W3].

A contaminated recording can still support descriptive findings. Missing electrodes, events or temporal support can make an endpoint unavailable without invalidating the whole recording. Plans may include downstream modules before prerequisite outputs exist, provided those prerequisites are included and run first. Later changes, recipes and selected processing inputs retain explicit versions.

Results distinguish completed, partial, unavailable, cancelled and software-error outcomes. A report or AI failure remains separate from a numerical failure. Processing history, execution attempts and evidence links allow the reader to establish what actually ran.

The restored rationale follows the original master [P0], compared with the retained Master Suite definitions and current release [M4, W5]. Historical and current method names are distinguished below.

The original method is a sequence of questions about an experience. Did the participant have access to the material? Did physiology track sensory structure, recognition, analytic work or artifact? Did an event coincide with a later change, and did that change persist? The analysis families preserve these questions at different timescales. Their names identify proposed interpretations of measurements; they do not make those interpretations established biological facts.

The logical chain begins with study identity, delivered stimulus, timing, channel identity and usable signal. It then compares candidate responses with sensory, structural, task, respiratory and order explanations. Event-level and continuous summaries follow those checks. Self-report, comprehension and later behavior constrain interpretation. A larger collection of favorable composites cannot compensate for missing timing or establish recognition, encoding or narrative reception without independent outcomes.

Primary questions, supporting explanations and exploratory candidates

The original master placed timing/QC, stimulus fingerprinting, artifact checks, A-MRED, MRED-Peak and MRED-Resolution in a proposed primary/core path. It placed access, confound and autonomic interpretation in a secondary layer, and KHT and Micro Handoff among exploratory candidates. These were intended study roles, not certificates of validation. Retained software registries also differ: HOC-R, OSA and OHC appear in the original secondary layer but in a later registry's exploratory layer. A study therefore needs its own frozen endpoint hierarchy, worker versions and eligibility rules.

The historical phrase “Target wins all” has only an endpoint-specific meaning. A strict A-MRED comparison asks whether particular Target recognition-like and delayed integration-like measures exceed their declared Control and Override comparators. It does not require Target to maximize every EEG, autonomic or subjective

variable. Override may increase task activity or preserve meaningful reception; Control may produce strong sensory responses; resolution may accompany decreasing activation. The direction of each prediction follows its scientific question.

ArtifactScore, GammaScalpel and the Temporal Semantic Proxy

ArtifactScore exists because eye movements, jaw tension, muscle, movement, electrical interference and unstable voltages can produce an apparent event. Its role is to restrict or qualify interpretation. Passing a composite artifact screen does not establish cortical origin, and unavailable artifact evidence must remain visible. Rejected windows and the duration retained after screening belong beside any resulting contrast.

The legacy GammaScalpel family decomposed lower-gamma activity and combined regional power, phase-locking and artifact sentinels into exploratory work-signal contrasts. Names such as MeaningGamma were operational labels, not measurements of meaning. The current Gamma Scalpel 2.0 and Measured Reference Companion add a distinct comparison of measured EEG with qualifying EMG/EOG evidence. These implementations must not be collapsed into one interchangeable score. Section 6.3 specifies the current screening method; an older result retains its original worker and calibration.

TSP means Temporal Semantic Proxy. Its original purpose was to locate time-resolved recognition/access-like changes that could be compared with narrative events. Retained implementations construct regional or feature contrasts. Electrode groups and residualized combinations do not identify a localized semantic transfer or isolate a semantic generator. TSP becomes informative through its relation to predeclared material, behavior and competing sensory explanations. Gamma and TSP often share inputs, so agreement between them is not independent confirmation.

MRED and A-MRED: recognition and later change

MRED means Meaning Recognition / Encoding Dissociation. It preserves the hypothesis that recognizing a familiar event can differ from acquiring or integrating something new. Its recognition-like component draws on gamma/TSP and related event measures; its encoding-like component uses later theta/carryover measures and, in retained versions, familiarity or novelty covariates. The labels describe a proposed distinction. Theta is not a direct measurement of successful memory encoding, and a familiarity rating does not independently validate a score that already incorporates it.

Four combinations remain useful hypotheses: strong recognition-like with strong later integration-like evidence; strong recognition-like with weak later evidence; weak recognition-like with strong later evidence; and weak evidence on both axes. A fifth outcome is unavailable or confounded. The third combination calls for checks of task, respiration, novelty, recovery and nonsemantic learning. The second can motivate a recognition or familiar-schema hypothesis, but does not demonstrate that no memory was formed. Later recall, recognition accuracy and transfer are separate measurements needed to test these interpretations.

A-MRED is the anchor-locked version. It asks a narrower question at a scene event whose identity, timing, windows and comparison rules were declared before acquisition. In its historical strict form, Target must satisfy recognition-like and delayed integration-like thresholds and exceed Control and Override by the specified margins after applicable quality and confound checks. Frame verification, content-to-recording alignment and anchor provenance are part of the evidence. A scene explanation supplied after inspecting physiology remains exploratory in that recording.

The delayed window is as consequential as the acute window. Choosing the most favorable delay after inspection changes the question. A window extending beyond a clip requires an explicitly declared continuation policy and adequate support; it cannot silently borrow another branch. A-MRED failure means that the operational endpoint did not pass. It does not establish that the scene was meaningless. A missing required component means the endpoint is ungradable, which differs from an observed negative result.

MRED-Peak and MRED-Resolution: impact and closure

MRED-Peak and MRED-Resolution were introduced because a single event shape cannot represent every proposed consequence of narrative experience. Peak emphasizes an acute, anchor-linked recognition-like response with delayed integration-like support. Resolution emphasizes a later reflective or regulatory pattern. Retained endpoint-compression software reads other analysis outputs and assembles interpretation tables; it does not supply an independent measurement of either construct.

The distinction allows a modest acute response to coexist with a later candidate for closure or recovery. It also prevents a strong acute response from being described automatically as sustained integration. Neither pattern should be selected retrospectively simply because it favors the hypothesis. Their windows, contrasts and independent behavioral predictions need prospective specification. A final reflection baseline is cumulative after-state: it may include prior branches, task relief, fatigue, respiratory settling and ordinary recovery. Its attribution is a separate analysis problem.

Confound defenses: what could produce the same pattern?

HOC-R, High-Order Control Residualization, asks whether faces, bodies, local shot content, biological motion or related structure explain an apparent narrative response. ShotOrderScramble strengthens this question by retaining more local structure than PhaseScrambling while altering sequence. Its preservation and disruption must be checked; neither control guarantees the absence of meaning.

OSA, Override Spatial Attention, asks whether viewers sampled an identical Target/Override film differently because they monitored cues, shifted gaze or squinted. File identity controls the media, not the participant's visual behavior. OHC, Order/Habituation/Carryover, addresses position, repetition, fatigue and preceding branches. A washout provides observations of recovery; its name does not prove that the participant reset.

AAM, Afterglow Attribution Model, asks which preceding state best accounts for a later baseline or washout. Target carryover competes with Override task-completion relief and mixed carryover. RespDualPath preserves respiration in two roles: physiological response and potential contributor to EEG or HRV findings. A sigh can be relevant physiology while also reducing confidence in a coincident electrical event. Adjusted and unadjusted analyses answer different questions and should retain their declared purposes.

CET-R, Cinematic Entrainment residualization, tests modeled contributions from luminance, audio envelope, motion, cuts and cue timing. Actual aligned stimulus regressors must be distinguished from a cue-only model or a file inventory. Residual variation is unexplained by the fitted model, not thereby identified as meaning. Missing regressors, model misspecification and fitted-on-the-same-data performance limit the interpretation. These defenses are complementary: residualization cannot replace physical timing, structural controls or measured artifact references.

NIP and EET: continuous patterns and later resemblance

NIP, Narrative Immersion Proxy, was intended to retain sustained candidate patterns that a single peak can miss. Its retained worker combines access-like and integration-like features with penalties. BIT, CII and IAQ provide threshold, continuous-density and attenuation views of this family. They reuse related measurements, so their joint appearance should be reported as several views of shared evidence. A high composite does not establish subjective immersion, and an attenuation ratio requires an interpretable, sufficiently supported denominator.

EET, Endogenous Echo Tracking, compares selected Target-event feature patterns with later washout or reflection patterns. Resemblance can motivate a persistence or reactivation hypothesis. It cannot identify what information persisted or show that the later pattern was caused by the Target event. Similarity depends on feature selection, normalization and available observations; ordinary recovery or a shared artifact can also

resemble an earlier pattern. EET therefore remains exploratory alongside null anchors and competing after-state references.

CAI/SID and Continuous Autonomic Arrays

CAI is the Controlled Access-Integration Index; SID is State Integration Density. The intended bridge is from isolated events to a supported trajectory: access-like activity, slower integration-like activity, positive-support terms and penalties are combined under a frozen specification. SID summarizes the accepted time represented by that trajectory. Their boundedness is a mathematical property, not a probability of awareness, meaning or memory formation.

The precise current specification and supported-time accounting appear in section 6.5 and Appendix E [M1]. Interpretation additionally requires knowing which components were available, how calibration was obtained, which future windows remained inside contiguous blocks and how much valid time remained. Future integration makes the full score retrospective. Positive product evidence and local correlation are different terms; a positive correlation alone does not demonstrate positive activation or a directional handoff. Contextual access ratings and autonomic outputs remain separately inspectable rather than silently changing the core measure.

CAA, Continuous Autonomic Arrays, restores when a change occurred. Beat intervals, heart-rate trends, rolling HRV, respiration and quality indicators can reveal an early response, sustained alteration or later recovery hidden by a branch mean. RR intervals are beat events, and rolling HRV represents a declared window rather than an instantaneous state. Breathing can account for part of an HRV change without making the breathing response irrelevant. Belt-derived depth is a relative measurement unless calibrated for another quantity. No single autonomic direction establishes narrative integration [M2].

Micro Handoff and KHT: timing and reciprocal association

Micro Handoff asks whether a brief rise in temporal gamma is followed by a rise in slower theta/integration-like activity. It distinguishes state agreement, positive directional change and activation-weighted density. Jointly low signals can resemble one another; stationary high activity can contribute density without a handoff. Its fixed lag bank and baseline calibration constrain selection. The 25 ms output grid does not provide 25 ms spectral resolution: overlapping feature windows and smoothing affect the timing that can be interpreted.

KHT examines reciprocal statistical relationships between a gamma/work signal and a human-translation proxy; event-local variants additionally ask about later theta carryover. Local association and carryover are separate requirements in that proposal. Reciprocal regression coefficients do not establish reciprocal causation, and their product's sign does not establish a biological feedback mechanism. Shared inputs, common drivers, temporal dependence and physiology-selected anchors require explicit checks. A candidate discovered in one recording supplies a hypothesis to lock for a later recording [M3].

Synthetic recovery tests show whether a declared planted pattern can be recovered under specified conditions. They do not establish the human construct. The historical Micro Handoff record includes mixed directional contrasts and failed null criteria; that history remains relevant even when later software tests pass. Both families remain exploratory unless a particular endpoint is prospectively evaluated against independent outcomes. Their value is in making timing, persistence and alternative explanations testable while preserving uncertainty about the proposed mechanism.

6.2 Methods and their intended questions

Method or family	What it examines
Spectral/aperiodic, ERP and time-frequency analysis	Declared frequency structure, event responses and supported time-frequency changes.
Connectivity/PAC, complexity, microstates and decoding	Statistical relationships or predictive features, with reference, artifact and evaluation controls.
MRED and A-MRED	Recognition-like versus later encoding/integration-like findings, with explicit anchor constraints for A-MRED.
Peak and Resolution views	Short responses versus sustained late patterns.
NIP: Narrative Immersion Proxy	Proposed immersion-related features; BIT, CII and IAQ provide threshold, density and attenuation comparisons.
CET: Cinematic Entrainment Tracking; EET: Endogenous Echo Tracking	Declared stimulus-related and continuing-response patterns.
NAST	Narrative-absorption proxy from reused signal features.
Legacy DGA	Sensory-access and subjective/contextual composite, distinct from the newer behavioral pilot.
CAI/SID	Operational access/integration-like feature combination and its supported-time summary.
Continuous Autonomic Arrays and RespDualPath	Beat intervals, trends, respiration, supported windows and separate physiological/artifact roles.
PPG/PRV and EDA review	Modality-specific processing and comparisons through named optional workers.
Micro Handoff and KHT	Directional changes, association, event timing and later persistence.

These names describe implemented measurements, not validated meaning, consciousness, DMN or memory detectors. Legacy DGA's overlapping ratings and missing-value defaults remain visible; its sigmoid is not a calibrated gate probability. Shared inputs do not provide independent confirmation. External EDF/BDF, EEGLAB and BrainVision imports preserve actual source identity, units, events and discontinuities [W3].

6.3 Gamma Scalpel: compare the artifact evidence

Gamma Scalpel 2.0 with Measured Reference Companion 2.1 examines power in `30 <= f < 45 Hz` using measured electrodes and native samples. It screens and compares recordings; it does not subtract contamination or certify cortical origin. Cranial muscle and ocular contributions motivate the comparison [G1, R4-R6].

The worker uses two-second nonoverlapping windows, linear detrending and Welch power with one-second subsegments, requiring EEG sampled at least at 100 Hz. Four views retain finite original windows, engineering-screened windows, EEG-proxy screening and measured EMG/EOG-reference screening. Sample continuity, coverage, amplitude, burst influence and accepted duration accompany the power estimates.

Reference qualification requires participant, acquisition, source, units, native rate and clock evidence. Unknown identity is not repaired by choosing a convenient trace. EMG burden uses native-rate filtering and RMS/envelope features; EOG uses low-frequency movement, amplitude and slew. These are incomplete

artifact indicators: recording-relative thresholds can miss sustained activity, and ordinary EOG screening is not complete microsaccade detection. Missing references remain unassessed.

The timing policy checks nominal and uncertainty-shifted reference windows. Its maximum **250 ms screening-uncertainty envelope** for two-second comparisons is an engineering rule, not samplewise alignment, submillisecond timing or phase qualification.

Condition matching uses measured artifact covariates on a common scale. Gamma is excluded from the matching distance. Matched duration is equal, with no window reuse **within each pairwise condition comparison**; different comparisons may share windows. Report unmatched duration and required-reference coverage because matching changes the moments represented. A reference-screened difference still needs a scientific interpretation.

6.4 Cleaning and preservation

Signal Quality & Cleaning Review keeps original, recipe preview, candidate and removed signal together, with annotations, source hashes and supported intervals. A reviewed candidate becomes a new input without replacing the raw recording or silently changing a plan.

Comparisons include spectrum interpolation, fixed-selection ZapLine, ICA review, paired reference regression and native SPAR-style candidates. Optional ICLabel is advisory; offline ASRpy requires a reviewed calibration interval. Reference regression is not iCanClean, and fixed ZapLine is not ZapLine-plus. ICA convergence and preprocessing/geometry requirements remain part of component review [G2].

The SPAR-style reference candidate is attributed to pinned upstream code, but authentic MATLAB numerical equivalence remains unestablished. Its muscle, ocular and slow passes can alter desired EEG. A separate preservation variant retains residual/support information. Conservative v2 leaves the slow pass unchanged unless reviewed, same-participant, timing-qualified independent motion evidence supports correction.

One retained 125 Hz single-channel synthetic clean control produced NMSE approximately **0.0837** for the reference candidate and **0.0291** for `preservation_v1`, against **zero** for unchanged input. The preservation variant also performed worse in some contaminated cases. Later policies retain separate identities; these adverse results do not disappear [G2].

Paired comparisons examine supported spectra, TFR, complexity, connectivity, gamma-theta and CAI/SID using common support and declared calibration. Reducing artifacts can also distort amplitude, lag, covariance or phase. Delayed processing currently qualifies recording replay, not physical live cleaning. Authentic reference outputs and held-out real preservation tests remain necessary.

6.5 CAI/SID and valid-time support

CAI is the **Controlled Access-Integration Index**; SID is **State Integration Density**. The frozen v0.2 core combines declared access-like, integration-like, positive-support and penalty features into a bounded unitless score [M1].

Read the normalization policy recorded for each result. The retained standalone core prefers Baseline 1 but permits declared full-run fallback and a standard-deviation scale when MAD is unusable. Its future interval is **8-30 seconds within the same continuous block**. The positive-amplitude product suppresses jointly below-midpoint contributions, while the worker's separate positive-correlation term can still contribute. This is a particular operational definition, not proof of integration; hard-invalid windows remain missing. Appendix E distinguishes these terms. Future support makes this an offline endpoint. Wholly missing NAS may be derived through NAST when coverage permits, with a separate enriched output and provenance. Partially observed NAS is not silently blended.

For accepted observations with nonoverlapping temporal support `dt_i`:

```
T_valid = sum(dt_i over accepted observations)
CAI_load = sum(CAI_i * dt_i over accepted observations)
SID = CAI_load / T_valid
occupancy(tau) = sum(dt_i with CAI_i >= tau) / T_valid
```

SID is unitless, load is score-seconds and occupancy is a supported-time fraction. Zero valid time makes the ratios unavailable. Overlaps and duplicate timestamps must not multiply duration, and nominal support must not bridge long gaps. The valid-time layer reports its own uncertainty status.

Paired processing regenerates its canonical features using original calibration and records differences from older fallback policies. Required montage, bandwidth, phase boundaries and baseline/future support must exist. Athena's absent canonical electrodes cannot be supplied by relabeling or reconstruction.

6.6 Physiology and temporal interpretation

Sampled heart rate does not become observed RR intervals or HRV through reciprocation. RMSSD uses only valid pairs adjacent in the original beat ledger:

```
RMSSD = sqrt(mean((RR[i+1] - RR[i])^2 over accepted original pairs))
```

Dropping an invalid interval before differencing can join nonadjacent beats. Gaps and connection boundaries therefore remain explicit; reviewed ectopic correction is a separate candidate. Raw interval variability is not automatically normal-to-normal HRV [M2].

Cardiac-respiration coupling requires participant/source pairing and suitable timing. PRAYCG's harmonic regression against measured breathing phase is distinct from optional NeuroKit RSA. PPG-derived PRV is distinct from ECG-derived HRV. Every trace should state its represented interval and processing support; a rolling estimate does not measure an instantaneous transition.

Micro Handoff separates state similarity, positive directional change and activation-weighted density, comparing ordered gamma/theta changes across a fixed lag bank. A **25 ms output grid is not 25 ms spectral resolution**. Different feature windows and smoothing contribute to fitted lags; scalp groups do not directly measure an STG-to-MTG transfer [M3].

KHT and related carryover analyses retain anchor and persistence rules. Reciprocal regressions do not establish reciprocal causation. Shared inputs, ordinary recovery and task relief require simpler alternatives and null-anchor comparisons. Synthetic lag recovery does not erase failed historical criteria.

6.7 Visual comparison

Live monitoring, replay and completed-analysis views retain their distinct sources. Paired TFR, gamma-theta and CAI/SID views preserve input identity, clock origin and declared calibration. Fixed scales and autoscaling should be visible.

Reporting and AI explanations follow the evidence requirements in section 10 [W3, W4].

7. Recorded studies and reconstruction results

These are selected exploratory recordings, generally with fixed condition order. They show measured differences and practical failures worth following up, but do not separate stimulus, instruction, familiarity, fatigue and elapsed time. Reanalyses remain attached to their original acquisition. This account summarizes saved outputs; no raw-data replay was performed for this edition.

Contact

The September 30 analysis retained 60 epochs per branch and evaluated 14 conditional timing scenarios. Its 75,600 rows represent repeated channel, band and scenario calculations, not independent observations. Posterior-temporal power differences were [D1]:

Comparison	Historical-map, zero-shift result	Range across timing scenarios
Target minus Control, theta 4-8 Hz	-0.419 dB	-1.310 to -0.325 dB
Target minus Control, beta 13-30 Hz	+2.826 dB	+2.469 to +2.826 dB
Target minus Control, gamma 30-40 Hz	+3.878 dB	+3.755 to +4.004 dB
Target minus Override, gamma 30-40 Hz	+4.844 dB	+4.600 to +5.212 dB

The ranges describe sensitivity to the selected timing assumptions; they are not confidence intervals. Gamma differences retained their direction, but uncertain montage/reference, eye and muscle activity, hardware artifacts and stimulus effects prevent attribution to a neural meaning process.

The evidence review recovered a cue file whose hash matched the recorded configuration. The conceptual anchor schedule was empty, so subsequently reconstructed anchors remain post hoc. A 39-cell ALS timing table was arithmetically reproducible, but three display pulses had been fitted to the same three branch markers. Those residuals did not independently validate physical synchronization.

Respiration timestamps indicated approximately 10 Hz despite a nominal 20 Hz label. An inappropriate ALS sample-index mapping compressed a roughly 2,216-second respiration span to roughly 178 seconds. Recovering the native span corrected that interpretation; it did not establish continuous valid respiration or cross-device alignment.

The CET-R stimulus-feature model used 904 rows and 13 features. Training R-squared was 0.0237; blocked cross-validation R-squared was -0.2608. The model predicted poorly on held-out blocks, so it cannot be used to declare stimulus confounding excluded.

Arrival

At zero shift, retained coverage was 63/63 Control, 62/63 Target and 57/63 Override epochs. All 30 region-by-band contrasts were estimable for each broad comparison under the stated coverage rule. Posterior-temporal gamma at 30-40 Hz was about +5.76 dB for Target minus Control and +11.68 dB for Target minus Override. The selected timing-scenario ranges were 5.39-6.61 dB and 11.24-11.68 dB [D2].

Artifact-sentinel gamma also differed, by about +3.47 dB and +5.57 dB. Thus the recording contains condition differences without isolating their neural origin. Missing source channel names and units leave physical mapping and reference uncertain. Agreement between embedded markers and an external event log supports event-record consistency, not electrode placement or physical stimulus delivery.

Field of Dreams and Her

Field of Dreams had one historical A-MRED anchor pass out of seven; an alternative content offset of +1.251 seconds reduced the count to zero. The ALS start check passed for one of three branches, and estimated anchors were not verified against delivered frames. This is a timing-sensitive observation rather than a stable event signature [D3].

Target ratings of Meaning 7, Absorption 7 and Afterglow 8 are recorded individual responses. They do not validate an EEG measure of those experiences. Later baseline activity could include narrative carryover, relief from Override, fatigue or respiratory settling.

Her produced a Target-local KHT-like candidate near 113.87 seconds. Its scene interpretation was supplied afterward. Delayed voice/lip synchronization, outside noise masking dialogue and blurry number cues also affected the run. A future repeat can freeze the proposed scene anchor and verify presentation before acquisition [D3].

Eternal Sunshine candidate

The archived candidate contains 240,000 samples across 16 EEG channels: 32 nominal minutes at 125 Hz. Yet the processed EEG timestamps span about 4,904 seconds and task markers about 2,024 seconds. The inventory records the film association as manifest-based inference, not independently confirmed file-to-film identity [D4].

Historical gamma contrasts include unavailable values; the broad theta and frequency-cascade criteria did not support the proposed pattern. Identity and timebase repair are required before a movie-specific physiological interpretation. A favorable legacy status label cannot replace a missing numerical estimate.

About Time and Snack Attack: coverage determines availability

About Time Run 1 was interrupted. Its 19,065 EEG samples cover about three timestamped minutes within an approximately 27-minute protocol; Target EEG is absent. Planned gamma and PNCC endpoints cannot be evaluated [D4].

The later review treated About Time Run 2 as a separate acquisition and applied a declared 80% branch-coverage requirement [D2]:

Recording	Retained Control / Target / Override epochs	Supported broad comparison
About Time Run 2	17/68, 63/68, 28/68	Neither full contrast met coverage
Snack Attack	9/55, 54/55, 55/55	Target minus Override; Target minus Control unavailable

Missing support is not a zero effect. Sparse reconstruction windows cannot replace the missing whole-branch evidence. Labels such as recognition-dominant or afterglow remain working descriptions; separate behavioral outcomes and prospective comparisons are needed to distinguish memory, relief, effort, arousal and artifact.

Zuna: prediction of withheld measurements

Zuna is an optional experimental reconstruction route. Withholding measured channels allows comparison with simpler predictions. Because the withheld measurements can contain artifacts, predictive accuracy is not a direct test of clean-brain recovery.

The September 30 offline comparisons used 12 fixed source positions, 36 model windows and 48 nominal scored seconds per withheld channel, or 16 seconds per condition. F3, F4, P3 and P4 were withheld; the

remaining 12 measured channels supplied inputs. Mean channel-pooled normalized mean squared error (NMSE, lower is better) was [Z1]:

Recording	Zuna	Spherical spline	Fixed zero	Combined accuracy gate
Contact	0.63037	2.79491	1.00036	Failed
Arrival	1.58457	1.06710	1.00209	Failed
Snack Attack	0.78158	0.88044	1.00013	Failed
About Time Run 2	1.02721	1.35634	1.00028	Failed

Zuna had the better average in Contact and Snack Attack. Arrival favored simpler alternatives. About Time Run 2 favored Zuna over spline but not fixed zero. None passed the retained combined requirements; favorable means coexist with failures at individual channels, windows or conditions.

Material limitations include template geometry, clipped coordinate encoding, source-clock and sample-rate assumptions, unknown physical reference and sparse coverage. Fifteen-second model contexts can overlap even when scored four-second windows do not. Changes in band-contrast direction after reconstruction show method sensitivity, not which direction is physiologically correct.

Earlier evaluations used different scoring scopes and are retained separately in Appendix D. The next test should freeze preparation and compare fresh recordings with simple baselines, while measuring preservation of events, spectra and channel relationships. Original measured data remain primary; reconstructed outputs remain identified derivatives.

8. Decoder Gate: design, predictions and results

Research question

Decoder Gate asks whether prior context and assigned instructions change access to a story, comprehension, later recall and physiological recovery in distinguishable ways. Interpretive context has experimental precedents [R1], and divided attention is an established explanation for memory differences [R2]. A useful test must distinguish the proposed prediction from attention, effort, sensory access, arousal and ordinary memory processes.

The original D223/D224 design compared intact narrative, scrambled material and intact narrative under arithmetic load. Those branches do not complete a structure-by-instruction factorial design. The new behavioral pilot adds the fourth combination and uses different stories within a participant's session [S2, S4]. Decoder Gate and OptoPING remain separate proposals: a behavioral interaction would not identify an optical feedback mechanism.

Two operational tests

The October 29 simulation tests whether context changes the physiological response to assigned surprise [S1]. For response `mu_j(c,s,tau)`, context `c` is binary, surprise `s` ranges from 0 to 1, `tau` is elapsed time after an event and `j` identifies the outcome. Holding nuisance inputs fixed:

```
Delta_j(tau) = [mu_j(1,1,tau) - mu_j(1,0,tau)]
             - [mu_j(0,1,tau) - mu_j(0,0,tau)]
             = theta_j(tau)
```

This asks whether context changes the difference between high- and low-surprise responses. The model allows either sign and summarizes the contrast over 4-10 seconds. Those are assumed feature-response times, not measured neural transfer times. A flexible additive competitor controls nonlinear surprise effects; the interaction model adds context-by-surprise terms.

The October 30 extension tests signal structure and assigned stance using independently scored comprehension and delayed memory [S3]. Let `S=1` denote coherent material and `S=0` reordered material; let `R=1` denote narrative instruction and `R=0` arithmetic instruction. For correct-answer probability `p_sr`:

```
Delta_behavior = (p_11 - p_01) - (p_10 - p_00)
```

This probability-scale contrast asks whether coherence has a different effect under the two instructions. With valid assignment and suitable participant/item analysis, it can estimate the total effect of those assignments. It is not interchangeable with a logistic-regression interaction coefficient.

A separate prediction test asks whether a fixed pre-probe physiological feature `P` improves prediction of answers beyond ordinary predictors, and whether its contribution varies by condition. The penalized logistic models are:

```
B0: condition cells; flexible measured attention and arousal;
    ability, item difficulty, movement and session-state proxies.
B1: B0 + P + P^2 + P^3.
B2: B1 + P*S_c + P*R_c + P*S_c*R_c,
    where S_c = 2*S - 1 and R_c = 2*R - 1.

loss(y,p) = -[y*ln(p) + (1-y)*ln(1-p)]
gain = mean(loss_simpler - loss_extended)
```

B0-to-B1 gain measures generic physiological prediction; B1-to-B2 gain measures the additional condition-dependent relationship. Physiology must be specified without using the answer it predicts. Adjusting for attention or arousal measured after assignment tests conditional prediction, not automatically a direct causal gate effect: those variables may be mediators or colliders.

Context-by-surprise simulation

The study evaluated 13 scenarios with 100 datasets each. Training, validation and held-out test runs were separate; scaling used training data and penalty selection used validation data. The illustrative screen required at least 2% relative test MSE improvement, an approximate paired-run interval above zero and coverage of all assigned combinations. Settings appear in Appendix B. [S1]

Assumed scenario	Mean prediction gain	Screen met
No relevant effect	-0.37%	0/100
Additive effects	-0.37%	0/100
Nonlinear additive effects	-0.37%	0/100
Context-dependent interaction	6.26%	69/100
Opposite-sign interaction	6.06%	69/100
Weak interaction	-0.20%	0/100
Interaction, fourfold noise SD	0.11%	2/100

Assumed scenario	Mean prediction gain	Screen met
Interaction, varying kinetics	8.15%	54/100
Identical nonsemantic pathway	6.26%	69/100
Unmeasured common causes	5.38%	49/100
Confounded training; context randomized at test	-5.75%	0/100
No shared context-surprise support	Not interpretable	100/100 withheld
Interaction, missing outcome samples	6.14%	65/100

The intended interaction was recoverable under favorable assumptions. An ordinary nonsemantic pathway produced exactly the same observations and results. An omitted common cause also passed in 49/100 datasets; supplying that cause exactly to both competitors removed the additional-interaction result. Such perfect measurement is an oracle comparison, not a practical guarantee.

The weak interaction existed but never met the screen. Conversely, zero detections in 100 null datasets does not establish a zero false-positive rate. The missing-outcome case retained known clocks and event inputs; it does not qualify reconstruction across lost acquisition intervals. These are synthetic operating characteristics, not probabilities that a biological theory is true.

Structure-by-stance behavioral simulation

The extension evaluated 11 scenarios with 100 datasets each. Whole participants and story families were separated across training, validation and test splits. The positive control assumed a shared state affecting both physiology and answers, with a stronger contribution in coherent/narrative trials. The screen required 0.005 nats per trial improvement and an approximate crossed participant/item bootstrap interval above zero. [S3]

Assumed scenario	Generic screen	Conditional screen	Conditional gain
Condition effects only	0/100	0/100	-0.0017
Physiology without behavioral link	0/100	0/100	-0.0019
Generic shared state	99/100	1/100	-0.0016
Condition-dependent additional state	27/100	6/100	0.0084
Weak additional state	0/100	0/100	-0.0015
Additional state, noisy physiology	2/100	0/100	approx. 0
Ordinary attention, noisy attention proxy	49/100	0/100	0.0002
Same world, true attention observed	0/100	0/100	-0.0006
Identical ordinary shared-state pathway	27/100	6/100	0.0084
Unequal physiological measurement quality	51/100	11/100	0.0112
Relationship reverses in held-out test split	0/100	0/100	-0.0346

Gains are mean primary comprehension improvements in nats per trial. **The specific test had poor sensitivity under these assumptions:** only 6/100 positive-control datasets passed, compared with 11/100 unequal-quality datasets. Secondary delayed-memory counts for those worlds were 4/100 and 14/100; these related outcomes are not independent replication.

Imperfect attention measurement often made physiology predictively useful without a new semantic mechanism. An ordinary shared-state explanation remained exactly indistinguishable from the proposed additional-state explanation. A reliability-informed comparison reduced unequal-quality detections to 0/100, but used known Gaussian reliability and does not establish that practical adjustment will remove the problem. The priority is better measurement and design, not relaxing the threshold after seeing results.

Laptop feasibility pilot

Reading, Attention and Memory is an unnormed behavioral pilot, with managed integration in Public Workbench Alpha 11.10.0 [S4, S5]. It runs without EEG, LSL or XDF. Optional physiology is a separate measurement layer.

Structure	Narrative instruction	Arithmetic instruction
Coherent	Follow the story; ignore digits	View the story and sum digits
Reordered	Follow the reordered story; ignore digits	View the reordered story and sum digits

Each story has four segments; reordering changes the first three while keeping the critical fourth segment last. Words, segment durations and digit schedules are preserved across versions of an item. Reordering and arithmetic can still alter difficulty, expectancy and comprehension strategy.

The bank has 32 main stories, two practice items, four calibration items and eight balanced assignment lists. A participant encounters each main story once across two 16-story acquisitions, with four stories per condition per acquisition. Items rotate through condition and immediate-fact form across lists. Changing a participant code does not remove prior exposure.

Immediate questions measure causal comprehension and one fact. Delayed follow-ups test both facts using a neutral story title. The outcome is title-cued factual recall; an immediately tested fact remains separate from one first tested at delay. The pilot has four conditions. It omits the simulation's four expectancy levels because no independently normed cue bank exists; its 32 stories are not the simulation's 96-item split bank.

Manual scoring uses frozen keys: 1 correct, 0 observed incorrect, U uncertain and M missing. A substantive "I do not know" is incorrect; a blank is missing. Accuracy includes scored, nonempty 0/1 answers and displays missing, uncertain and unscored counts alongside its denominator. Original answers and scored/adjudicated copies remain separate. Scorers should be blind to condition and physiology where feasible. Participants should not inspect materials or keys before completing follow-ups.

Decision rules and next use

The pilot first tests reading pace, answerability, scoring agreement, arithmetic burden, adherence and sensor reliability. Missing physiological support does not invalidate an otherwise usable behavioral answer. Failed arithmetic or inconvenient comprehension trials must not be removed to favor the hypothesis. Revised materials require a new version; exposed participants or items are not untouched evaluation data.

A later evaluation needs a prospectively chosen endpoint, meaningful effect scale, support rule and uncertainty method. Strong baselines matching or exceeding extended models challenge incremental prediction. For a prespecified meaningful effect `delta`, an appropriately constructed interval wholly within `[-delta, +delta]` supports a bounded negligible-effect conclusion; a wide interval is inconclusive. A

directional prediction must also be frozen before evaluation. Missing condition coverage makes the comparison unavailable. Generalization requires fresh sessions, stimuli or participants.

Neither an interaction nor incremental prediction uniquely identifies a semantic gate. Observationally identical mechanisms require new discriminating measurements or interventions. Recognition, comprehension, recall, subjective impact and recovery remain separate outcomes. The simulations and software tests establish no human effect and authorize no live protocol adaptation.

9. OptoPING: physical feasibility and discriminating tests

The original OptoPING proposal is broader than photon-receptor feedback. It asks whether an intracellular structural mode could influence membrane excitability and participate in reciprocal coupling with electrical activity. The optional reconciliation appendix permits an optical, electrical, vibrational or mixed mode. Its proposed bridge is a cytosolic transfer process that converts that mode into a change in membrane threshold or forcing. Neither the carrier nor the transfer gain has been established by the equations [T10].

The meaning of `Y` therefore needs to remain explicit. D274 distinguishes an equation-level state, a simulated state, an inferred human transition state, a measured cellular proxy and a memory-relevant structural variable that survives competing explanations. These are different evidential claims. A fitted latent state in scalp EEG does not become a measured cellular mode, and a synthetic recovery does not identify the substrate [T11].

The early Rung 0 simulations test whether stipulated reciprocal couplings can be recovered from dynamical observations. They do not implement intracellular topology or the protocol's separate hypothesis that structural or regulatory refresh preserves a pattern through molecular turnover. Later feature and score tests address narrower discrimination questions under their declared generators; Appendix C.3 separates these stages [T10, L1].

The October 24 model takes a different step: it specifies a possible photon-sensitive receiver and asks whether its required local photon supply and response dynamics are physically plausible. Its receiver occupancy is not interchangeable with the early oscillatory `Y`. This is one candidate pathway within the broader proposal. Connecting the models requires measured source emission, transport, receiver sensitivity and coupling to electrical activity; their shared label does not supply that evidence [O6].

Persistent neural activity also has ordinary synaptic, circuit and metabolic explanations. Each proposed bridge must earn its place by predictions that those alternatives do not explain. The following photon-specific account retains the October 24 assumptions and results; it is not a microscopic QED derivation or a gamma-generating PING simulation [O6].

The light supply required by a receiver

Let `y` be the fraction of receivers in an active state, `Phi` the local photon fluence rate within the effective activation band, `sigma` the absorption cross section and `q` the photoconversion probability. With activation rate `k` and deactivation rate `d = 1/tau_receiver`, the simple receiver model is:

```
dy/dt = k (1 - y) - d y
k = q sigma Phi
y_equilibrium = k / (k + d)
Phi_required = y_equilibrium /
               [(1 - y_equilibrium) q sigma tau_receiver]
```

This connects a proposed active fraction to the light required to maintain it. It does not tell us which fraction would affect a neuron; that relationship needs independent measurement of the receiver and its downstream response.

The reference calculation assumes 100,000 receivers, `sigma = 2 x 10^-16 cm^2`, `q = 1` and a ten-second effective dark lifetime. None is established here for native cortical OPN5. The illustrative target is 1% occupancy, not a physiological detection threshold. Under these assumptions, with no reverse photoconversion:

Local activation-band fluence, photons/cm^2/s	Unsaturated activation-count scale over 10 s	Equilibrium active fraction
`10^3`	`2 x 10^-7`	approximately `2 x 10^-12`
`10^6`	`0.0002`	approximately `2 x 10^-9`
`10^10`	`2`	approximately `0.000020`
`10^12`	`200`	approximately `0.001996`
`5.05 x 10^12`	`1,010`	`0.01`

The count scale is `N q sigma Phi T`. It is not the number active at the endpoint: receivers deactivate and may activate again. Starting with inactive receivers, the reference fluence produces an exact expected active population of approximately 636 after ten seconds, rather than the equilibrium population of 1,000.

The distinction matters at low fluence. An expectation far below one activation is not a small activation delivered reliably to every cell. The simulation uses finite-state transition probabilities rather than turning fractional expected photon counts into guaranteed events. At high fluence, reverse photoconversion can also limit occupancy. An assumed stress case in Appendix C makes the 1% target unreachable; it is a test of the model's behavior, not measured OPN5 kinetics.

What biological observations establish

Neural preparations emit ultraweak light. Isojima and colleagues measured approximately `10^-19 W/mm^2` from rat hippocampal slices; potassium depolarization increased emission and tetrodotoxin reduced it [O1]. Those observations support a relationship with neural activity and metabolism. They do not show that the escaping photons activated a receiving neuron.

Assigning all that external power to 380-nm photons would give approximately 19 photons/cm^2/s at the measurement surface. This is a conditional unit conversion, not a measured emission spectrum. It cannot be divided directly into the receiver budget above to obtain a defensible biological "photon deficit." Escaped light at a detector and illumination at a nearby receptor are different quantities. Collection geometry, absorption, escape fraction and source-receptor distance must connect them.

Kobayashi and colleagues observed brain emission associated with EEG activity, blood flow and oxidative conditions; glucose removal and mitochondrial inhibition changed it [O2]. This makes ordinary oxidative chemistry a serious explanation for changing light output. More emission need not mean more useful signaling. A proposed communication mechanism must explain what the photons do after production.

OPN5 provides a biologically grounded receiver candidate. Kojima and colleagues established near-380-nm sensitivity, Gi signaling and bistable photochemistry in reconstituted mammalian OPN5 [O3]. That preparation does not establish the abundance, absorption cross section, effective lifetime or feedback gain required by the cortical model. A persistent photoproduct is a candidate state variable, not a measured duration for the complete native pathway.

Zhang and colleagues demonstrated OPN5-dependent responses to external violet illumination in mouse hypothalamic neurons [O4]. Their attenuation measurements and sunlight extrapolation yielded a possible

local violet flux of approximately `9 x 10^12 photons/cm^2/s`. This estimate concerns externally supplied light. It is neither an endogenous emission measurement nor a human cortical response threshold. Native response in that preparation supports investigating the candidate; it does not establish the proposed cortical source-receiver arrangement.

Tang and Dai's glutamate-evoked slice luminescence supplies another observation to examine [O5]. Their experiments included prolonged exposure to 50 mM glutamate and a gradual buildup over approximately 90 minutes. That preparation and time course do not establish physiological millisecond feedback or a naturally colocalized emitter-OPN5 pair.

Together, these studies justify investigating light production and reception. The missing link is quantitative: whether ordinary activity supplies enough photons in the appropriate band, at the appropriate location, to drive an independently measured receiver response and alter neural activity.

What the simulations can distinguish

The physical-feasibility study compared optical, metabolic, synaptic, mixed and chemically confounded mechanisms under several perturbations [O6]. Under its assumed optical mechanism, strong photon-path suppression reduced the modeled activity and matched rescue restored it close to sham. The measured endpoint was activity during stimulation, not after-state persistence or EEG voltage. Simulation uncertainty describes repeated noise realizations at fixed parameters, not uncertainty about the biology.

The decisive result was the competing explanation. Optical and metabolic mechanisms could produce exactly the same training and sham trajectories. They separated under blockade only when the intervention was assumed to affect the optical pathway selectively. When a chemical alternative was allowed corresponding effects of both blocker and rescue, it reproduced the optical trajectories exactly. A convincing block-and-rescue pattern is therefore insufficient when the manipulations also change the competing chemical pathway.

Good prediction also failed to identify every parameter. Different activation-rate and feedback-gain combinations fitted the observations. The Rung 0 trajectory, feature and score exercises test different versions of recoverability under stipulated generators. Their favorable results are not biological sensitivity or specificity. Appendix C.3 now recovers that distinction and retains the null-test failures.

Tests that could reject a specified pathway

A useful optical hypothesis must name its source, receiver, wavelengths, location and predicted effect size. The effective activation rate can then be bounded using measured spectral quantities:

```
k_opt = integral[q(lambda) sigma(lambda) Phi(lambda) d(lambda)]
```

Here `Phi(lambda)` is local spectral photon fluence rate per unit wavelength, with matching wavelength units throughout the integral. Compare its supported upper bound with the minimum activation required by independently measured receiver kinetics and downstream gain. If even that upper supply bound cannot produce the proposed effect within those measured constraints, the specified model fails. An assumed gain or arbitrary 1% occupancy target cannot provide the independent minimum.

A second test concerns pathway dependence. A sufficiently strong, verified interruption of the proposed optical path should change the declared endpoint. If the endpoint remains unchanged within a prospectively specified informative interval, the experiment bounds or rejects that contribution for the preparation. Receiver-loss tests require consideration of compensation; optical blockade and rescue require chemical, electrical and thermal controls. A broad interval remains inconclusive.

Finally, compare predictions on perturbations not used to fit the model. If ordinary synaptic or metabolic alternatives predict those observations equally well, the available evidence does not identify an optical contribution. Near-field transfer, freely propagating photons and redox signaling require separate models. More measurements of a trajectory that all alternatives predict identically cannot resolve which mechanism produced it.

10. Research workflow, reporting and exchange

A traceable study

Public Alpha 11.11.0 retains managed compact Cerelog acquisition and the behavioral Decoder Gate v1 pilot. It adds a background saved-recording inventory, advisory prepared-study review, shared review cursors, paired TFR displays, per-channel preservation summaries and clearer execution/measurement/evidence reporting. It also repairs current catalog and protocol-export release bindings. The release's automated and native-interface checks concern software behavior; physical qualification remains separately scoped [W5].

The prepared-study service distinguishes declared people, sessions, story families and trials; missing identities remain unknown. Its comparison route uses common held-out observations and training-only fitting against declared ordinary alternatives. These tools help expose design errors and unsupported claims. They do not choose a theory's endpoint or turn an exploratory run into a prospective study.

An acquisition record should identify the exact protocol and material, equipment and channels, participant/source pairings, events, timing evidence and deviations. Preserve raw data; make processing and exclusions versioned derivatives. Each result should identify its inputs, settings, method version, accepted intervals, annotation version and execution state. Record what was actually acquired or executed rather than infer it from the intended plan.

Readable results with accessible evidence

The opening of a module report should answer six questions: what was examined, what was found, what the measurements mean, which comparisons are supported, what remains uncertain or unavailable, and what would be useful next. Tables, definitions and source evidence remain accessible underneath. A contaminated recording can still support descriptive findings, as the case accounts in section 7 demonstrate.

The PRAYCG AI Assistant explains completed outputs. Numerical workers calculate statistics; an explanation must cite those outputs, retain failed and unavailable endpoints, and distinguish observations from interpretations and hypotheses. A consolidated report should identify where findings reuse the same data or component scores. Report-generation failure must not erase the underlying analysis. Explanations and human revisions are separate, versioned derivatives; existing results can receive explanations without rerunning analyses or changing the plan [W4].

Plots should identify source data, processing, units, scales and represented intervals. Live observation, recorded replay, processed derivatives and reconstruction are distinct views. A visually convincing trace is useful for inspection, but the endpoint's method and support determine the claim.

AI-assisted authoring

The protocol builder packages the question, hardware constraints, proposed stimuli, outcomes and a return contract for external AI review. Intake reviews the returned proposal against supported task logic, configuration, materials, events, termination behavior and analysis requirements. The same process applies to recipes: retain units, dependencies, numerical examples, method references and resource limits.

Current developerless intake supports data-only protocol components and recipe changes within existing methods. A new estimator, connector or executable runner still requires implementation and dedicated testing. Returned packages remain inert during review; approval applies to an identified source and configuration. A changed endpoint needs a new version [W4].

Bundles, privacy and reuse

Research bundles can preserve protocols, stimuli, hardware profiles, data, events, plans, recipes, reports, environment descriptions and a file ledger. The local PRAYCG DATA catalog records relationships between original studies, reanalyses and replications. Independent reopening and execution must be tested separately from checksum verification [W2-W4].

An optional full-telemetry AI-review bundle may include XDFs, channel maps, anchors, familiarity reports, written answers and scoring keys. These contents may identify a participant without an explicit name. Public selection therefore requires a review of consent, metadata, media rights and the actual files leaving the computer. Imported bundles do not automatically install code, overwrite the source study or authorize publication. Specialized pilot bundles retain their own schema and tested compatibility.

Method citations credit scientific ideas; redistribution notices preserve applicable software rights. Record the actual implementations and observed versions used. Keep upstream license texts where required and distinguish shipped artifacts from optional downloads. Neither PRAYCG's license nor this bibliography grants rights over upstream components. Historical run identities and numerical outputs should survive citation or display-name changes.

Human studies require consent, disclosure of relevant stimulus content, privacy review and a workable stop/withdrawal procedure. A device's functional test is not electrical-safety certification. The Workbench is research software, not a diagnostic or treatment device.

11. Research milestones and decision criteria

Restore the theory without delaying the next useful recording

Keep a versioned distinction between the cellular storage hypothesis, its proposed bridges and the human state/behavior predictions. For each study, select one claim and its simplest viable alternative before choosing a composite score. The reconstructed equations are candidates for review; they do not create new measurements in old recordings.

For Topo-OSM's human track, cue-specific reactivation remains the most direct near-term test of the preserved intent. Compare appropriate cues with wrong-story cues and matched sensory controls, independently score recognition and delayed retention or transfer, and freeze the physiological feature and scoring procedure before evaluation. A physiological resemblance alone cannot identify memory content. Compare prediction with stimulus history, familiarity, attention and ordinary recovery on held-out sessions or items.

A prospective extension can add a small, fixed set of before-and-after probability questions about the same outcomes, then score later observations without changing the scoring key. Keep these updating endpoints separate from relevance, recognition and delayed recall. Additional questions can themselves change attention and memory, so the extension needs its own version and planned placement. It must not be inserted retrospectively into a frozen pilot or inferred from an older recording that lacks the required reports.

The standalone cue-reactivation materials and later recognition/confidence calibration work retain their own versions. At the Public 11.11.0 release cutoff, a qualified cue-reactivation presenter/analysis intake and the

newer DGA/CCP revisions had not been adopted into the Workbench. Documenting a protocol is distinct from qualifying its presenter and scoring workflow [W5].

The first decision is whether these measures are answerable, reliable and sufficiently precise to distinguish the specified predictions. Freeze a primary contrast, a scientifically meaningful effect scale, support rules and uncertainty method before an evaluative run. A confidence interval inside the prespecified negligible-effect region can challenge that prediction; a broad interval calls for better measurement or more data. Interpersonal alignment is a separate optional extension, not a prerequisite for the individual test. A stronger cellular claim requires a directly measured storage variable and selective perturbation evidence in a suitable laboratory preparation.

First acquisition priority: complete a qualified PRAYCG4 run

1. **Qualify the combined sensors.** Use the compact Cerelog configuration with Cyton/Daisy, both ALS channels, Polar and Vernier. Document physical placement, units, contact behavior, source pairing, gaps, recovery and electrical-use limits. Fit timing on calibration events and assess separate held-out optical events. Deliver a qualification record specifying which channels and endpoints have adequate support.
2. **Freeze the study package.** Identify and fingerprint every branch, review control audio and task cues, and register condition order, anchors, washouts, outcomes and the intended meaningful-effect criterion. Keep calibration and development recordings separate from evaluation data.
3. **Record complete basement and shielded sessions.** Preserve raw XDFs, events, channel maps, timing receipts, deviations and behavioral reports. The environments are distinct conditions, but session, familiarity, contact and order can also differ. Attribution to shielding requires a design that addresses those alternatives.
4. **Evaluate gamma and signal preservation.** Compare original, proxy-screened and measured-reference Gamma Scalpel views on declared support. Inspect the moments removed by matching. Evaluate cleaner-induced distortion as well as artifact reduction, with no-change outcomes retained. Obtain reference outputs before claiming numerical equivalence for a port.
5. **Complete the research record.** Run the planned analyses, explain supported findings and failures, verify the bundle, and reopen it with another machine or operator. A successful endpoint should survive its declared timing, coverage and confound tests before motivating a stronger claim.

Progression from individual records to prospective studies

First establish that the qualified pipeline can be repeated across sessions in one individual, retaining failed and incomplete runs. Then test feasibility across people and story families: instruction comprehension, adherence, signal coverage, timing, tolerability and endpoint availability. Those observations inform a separately frozen prospective group study, with declared primary endpoints, sample-size rationale, assignment and analysis rules. Repeated clips or windows within one person do not replace independent participants. Development data should not also serve as the held-out test of a revised score. Replication across sessions, materials, participants and operators answers different generalization questions [P0].

Parallel studies

Decoder Gate: establish reading pace, answerability, scorer agreement, task burden and delayed-follow-up feasibility. Measure condition-dependent signal reliability before using physiology as a predictor. Norm any expectancy manipulation separately. The synthetic conditional-positive detection rate of 6/100 calls for improved sensitivity and design, followed by a frozen evaluation on untouched participants/items.

Reconstruction: freeze Zuna preprocessing, geometry, channel selection and decision criteria, then compare fresh cases with simple baselines. Withheld-channel prediction, artifact attenuation and preservation of desired features are separate tests. Original measured data remain the reference record.

OSM-CELL and OptoPING: specify the preparation, measured structural variable, transfer route and ordinary synaptic/metabolic alternative. First establish the proposed encoding-dependent tuning or coupling; then test its persistence and retrieval contribution. The photon-budget and selective-intervention tests in chapter 9 apply to the optical candidate. Direct cellular work requires a separate qualified laboratory collaboration.

Independent accessibility and reproducibility: document cost, setup effort, failed installation or acquisition paths, and independent hardware builds. Test whether the workflow prevents known research errors and whether another operator can reproduce a declared result. These outcomes make the accessibility claim measurable.

Later adaptive protocols

Begin with passive recommendations and deterministic replay. Before live adaptation, define allowed actions, delay, resource bounds, consent, stop/recovery behavior and an experimental comparison that can identify the intervention's effect. The system should retain what it observed, recommended and changed. A model's interpretation of incoming telemetry is not itself evidence that the intended psychological state occurred.

A browser or mobile interface may improve access later. Hardware acquisition and physical timing will still require a qualified local service. This expansion should follow reliable recording, analysis and independent reopening of the current studies.

Other theory families

Gate optionality, precision-weighted updating and causal/extropic proposals require their own measured variables and competing predictions before becoming scientific endpoints. A theory archive can preserve them without presenting every proposal as a validated protocol. Thermodynamic load hypotheses require calibrated metabolic measurements: EEG and HRV alone do not supply joules, ATP or irreversible dissipation.

12. What v3 preserves and changes

The experimental account restored

The v1.1 master explained PRAYCG through its experimental choices. This edition restores that account: the full PRAYCG3 sequence, branch meanings and limits, input matching, report timing, familiarity, anchor selection, washout and final reflection. PRAYCG4 and SMG explain how the program extends its original question. Self-report is retained as a distinct kind of observation, with possible agreement and disagreement with physiology.

Recognition without a large new update, acute impact without prolonged after-state, and quieter resolution with later persistence remain candidate patterns. They are useful hypotheses because they can motivate different measurements and failures. They are not established categories or labels assigned solely to preserve a favored explanation of a run.

Corrections retained or added

The mathematical repairs in v2.4 remain in Appendix A. Meaning is not reduced to belief divergence; interpersonal update alignment is not identified with empathy; persistent amplitude is not identified with retained content. The optical, cellular and human-scale accounts retain different evidential requirements.

The original v1.1 SID presentation equated a mean over valid observations with division by the full interval. V3 retains the valid-duration denominator, and Appendix E shows why missing support changes that equivalence. Original beat adjacency is retained for RMSSD. Temporal feature windows remain distinct from the displayed sampling grid.

One additional implementation clarification emerged during this merge. The positive-product component of CAI excludes joint below-midpoint activation, but a separate positive-correlation term can remain nonzero. The earlier sentence that jointly low signals never count was broader than the worker's definition. V3 corrects that description without changing code or old results. The exact version, component weights and time support remain essential to interpreting any score.

The original all-positive Target contrast is treated as a prediction for an explicitly named endpoint. Different endpoints can predict transient, delayed or absent responses. A universal rule that every measure must favor Target would erase the recognition/integration and peak/resolution distinctions that motivated the project.

What moved and what remains historical

Software release chronology, exact CSV layouts and installation steps belong in their versioned release documentation. Their research functions remain here: traceable identity, explicit missingness, reproducible timing, transparent method definitions and inspectable outputs. Earlier workflows, formula versions and reports remain accessible as historical records; they are not rewritten into the current contract.

Appendix F maps all 36 numbered v1.1 sections to their treatment in v3. No omission should imply that a question was disproved or that a historical specification is current. Source identities and the accompanying revision ledger allow the comparison to be checked.

Appendix A. Reconstructed mathematical specification

This appendix preserves the proposed encoding-storage-retrieval chain: activity changes a structural response, and a later cue preferentially reactivates the stored pattern. It separates that hypothesis from its possible membrane/network bridge and from human EEG measurements. The archived equations are specifications to investigate, not fitted descriptions of a demonstrated biological memory substrate [T1-T5].

Status of the reconstruction. The bounded phosphorylation equation and resonance hypothesis retain the archived architecture. The calcium monotonicity condition, oscillator peak, phase-locking statement and coupling conventions repair mathematical claims. The bounded refresh example and its stability calculation are explicitly **new formalizations**, offered to clarify the original turnover question. They are not recovered experimental findings or changes to Workbench analysis workers.

A.1 Quantities, units and encoding

Time `t` is in seconds. Membrane voltage `V` and calcium reversal potential `E_Ca` are in volts; conductance `g` is in siemens; inward calcium current `I_Ca` is in amperes. Calcium concentration `C` and its baseline `C_0` use the same declared concentration unit. Phosphorylation fraction `P` lies between zero and one. Angular frequencies and decay rates have units `s^-1`, with radians treated as dimensionless.

The network patch makes encoding depend on integrated excitation, inhibition and plasticity gain [T2]. A dimensionally specified version is:

```
V(t) = V_rest + integral_0^infinity h_m(r) I_net(t-r) dr
I_net = I_exc - I_inh + I_other
g_eff(t) = G_N(t) g_Ca_NMDA,              G_N(t) >= 0
I_Ca(t) = g_eff(t) s(t) B(V(t)) [E_Ca - V(t)]
```

Here `h_m` has units `V/(A s)`; `s` and magnesium-unblock factor `B` are dimensionless fractions. `g_Ca_NMDA` is a calibrated, nonnegative effective calcium-component conductance, not the total mixed-ion NMDA conductance. The influx expression is restricted to an inward-current regime, `V < E_Ca`. The calcium state equation below uses nonnegative influx. A model covering current reversal requires the corresponding ionic balances rather than silently using a negative concentration source.

`G_N` is an effective gain, potentially influenced by novelty, inhibition and neuromodulatory state. Calling it a novelty gate does not establish semantic specificity or imply that neuromodulators universally multiply NMDA conductance. Its physiological implementation and nonnegative parameter domain must be specified.

The archived claim that higher voltage necessarily increases influx needs an additional condition. Holding `g_eff` and `s` fixed:

```
dI_Ca/dV = g_eff s [B'(V)(E_Ca - V) - B(V)]
dI_Ca/dV >= 0  iff  B'(V)(E_Ca - V) >= B(V)
```

The equivalence assumes `g_eff s > 0`. Increasing unblock competes with decreasing driving force. Monotonic `B` alone is insufficient; changing receptor opening or gain adds further terms. Familiar-versus-novel inequalities therefore need measured conditions or explicitly restricted assumptions.

A bounded encoding model is:

```
dC/dt = -(C - C_0)/tau_C + alpha I_Ca
H(C) = C^n / (K_C^n + C^n)
dP/dt = k_ph H(C)(1-P) - k_deph P
```

Require `tau_C > 0`, `K_C > 0`, `n > 0`, `C_0 >= 0`, `alpha >= 0`, nonnegative rates and input, nonnegative initial concentration and `P(0)` in `[0,1]`. `alpha` has concentration units per coulomb; `k_ph` and `k_deph` are rates. Then

concentration remains nonnegative. At `P=0`, its derivative is nonnegative; at `P=1`, it is nonpositive. Thus the continuous equation preserves `0 <= P <= 1`. A numerical solver must also respect this domain. Net `dP/dt` can be negative during dephosphorylation; it is not interchangeable with the positive encoding drive `k_ph H(C)(1-P)`.

A.2 Stored tuning and selective retrieval

The original storage hypothesis is that biochemical state changes a measurable response function [T1,T2]:

```
Omega_m(P) = Omega_0 + kappa_P P + O(P^2)
```

This is a local approximation, with `kappa_P` in `s^-1` per unit fraction and positive `Omega_m` over its intended domain. It does not establish whether the relevant mode is optical, electrical, mechanical or mixed. A mode-specific experiment must measure the change, its duration and its dependence on the proposed encoding process.

For a fixed stored state, consider the second-order driven oscillator:

```
x'' + Gamma x' + Omega_m^2 x = F cos(Omega t)
A(Omega) = abs(F) / sqrt[(Omega_m^2-Omega^2)^2 + Gamma^2 Omega^2]
```

`x` has a declared response unit, `F` that unit per second squared, and `Gamma > 0`. The expression is the steady-state amplitude under constant forcing amplitude. A first-order complex optical-amplitude equation is a different representation; equivalence requires an explicit approximation and normalization.

For nonzero forcing, the exact amplitude peak is:

```
Omega_peak = sqrt(Omega_m^2 - Gamma^2/2), if Gamma^2 < 2 Omega_m^2
Omega_peak = 0,                           otherwise
A(Omega_m) = abs(F)/(Gamma Omega_m)
```

Consequently, the peak is near the undamped natural frequency only with sufficiently weak damping. Near resonance, when `abs(Delta)` and `Gamma` are small relative to `Omega_m`:

```
Delta = Omega - Omega_m
A(Delta) approximately abs(F)/(Omega_m sqrt(4 Delta^2 + Gamma^2))
A(Delta) > A_star requires:
abs(Delta) < 0.5 sqrt[(abs(F)/(A_star Omega_m))^2 - Gamma^2]
```

The bandwidth statement requires `A_star > 0`, a positive radicand and belongs to the approximation. If `F=0`, every frequency gives zero amplitude. A biological recall threshold `A_star` is an additional hypothesis, not a consequence of oscillator mechanics.

One scalar can represent only one tunable degree of freedom. The distributed extension instead uses a pattern of states and response functions across specified sites [T2]. A matching cue may activate enough of that pattern to distinguish it from alternatives. Demonstrating stronger response at a matched frequency would establish selective physical response; establishing memory additionally requires encoding history, cue specificity, behavioral relevance and controls for ordinary sensory tuning. The equations alone do not assign autobiographical content to a resonance.

A.3 Maintenance under molecular turnover

The refresh proposal asks how an information-bearing organization could survive replacement of its material components [T3]. Its state vector is:

```
X_mem = [P_tr, B, W, Z, R]
```

P_tr denotes structural trace strength, distinguished here from phosphorylation fraction P; their relationship must be specified. B denotes scaffold support, W circuit organization, Z regulatory support and R reactivation. These are distinct candidate variables, not five already measured components of a memory. Regulatory support does not mean a complete memory encoded as a DNA file.

The archived refresh law includes saturation, but unrestricted additive disturbances can drive a bounded trace negative. Its linear scaffold correction can also make loss negative. The following **new bounded realization** retains the proposed competition between maintenance and loss:

```
p = P_tr/P_max,                          0 <= p <= 1
dp/dt = a(p,B,W,Z,R,t)(1-p) - b(B,t)p
a >= 0,                                  b >= 0
```

Both rates have units s^-1. For rates continuous in time and locally Lipschitz in the state, finite over each observation interval, the stated nonnegative boundary rates preserve the interval. One possible scaffold-dependent loss is b=lambda_0/(1+rho B) for nonnegative, normalized B and rho; this illustrative choice is not an identified biological law. Fluctuations require a process that preserves the domain, rather than arbitrary additive noise followed by unreported clipping.

A scalar example isolates what a refresh threshold can and cannot mean:

```
dp/dt = (a_ext + k_s p)(1-p) - lambda p
a_ext >= 0, k_s >= 0, lambda > 0
```

With a_ext=0, a positive equilibrium p*=1-lambda/k_s exists when k_s>lambda, and is locally stable. For k_s<lambda, zero is stable and positive traces decay. At equality, decay is algebraic rather than exponential. Exact zero remains zero without a seed even when the positive equilibrium exists.

That threshold is specific to self-dependent renewal in this example. With k_s=0 and sustained external support, p*=a_ext/(a_ext+lambda)>0 for every positive a_ext, including a_ext/lambda<1. Therefore the archived aggregate ratio cannot universally imply that every subunit ratio entails complete trace loss. A positive driven level also does not establish a self-sustaining memory.

For the full coupled system, a proposed equilibrium must satisfy all state equations. At an equilibrium of a continuously differentiable autonomous system, a Jacobian with eigenvalues having negative real parts gives local asymptotic stability; it does not prove global attraction or preservation of information. Instantaneous eigenvalues are not sufficient for a time-varying system. Trace identity requires a separately defined pattern-distance or retrieval criterion. A scalar amount can persist while its encoded pattern changes. The public refresh note's shorthand adds copy error and noise before dividing by a correction gain [T12]. That expression does not specify whether noise is a drift, amplitude or variance. It cannot supply a general fidelity threshold without a stochastic model.

New formalization for local fidelity error. Let e_f be one signed, calibrated error coordinate relative to a specified target pattern. It has a declared unit U. For a fixed target and constant coefficients, one illustrative correction process is:

```
de_f = (b_f - k_f e_f) dt + sigma_f dW_t,        k_f > 0
E[e_f(t)] = e_f(0) exp(-k_f t)
           + (b_f/k_f)[1-exp(-k_f t)]
Var[e_f(t)] = sigma_f^2[1-exp(-2 k_f t)]/(2 k_f)
E_stationary[e_f] = b_f/k_f
Var_stationary[e_f] = sigma_f^2/(2 k_f)
MSE_stationary = (b_f/k_f)^2 + sigma_f^2/(2 k_f)
```

The finite-time variance assumes a deterministic initial error, with sigma_f >= 0 by convention. W_t is a standard Wiener process. k_f has units s^-1, b_f has units U/s, and sigma_f has units U/sqrt(s). Both terms in the mean squared error have units U^2. A nonzero mean error from systematic drift is distinct from variance

caused by stochastic disturbances. The stationary identities follow by setting the mean equation m'=b_f-k_f m and variance equation v'=-2 k_f v+sigma_f^2 to zero.

For sigma_f>0, the stationary distribution is Gaussian with mean mu=b_f/k_f and standard deviation s_f=sigma_f/sqrt(2 k_f). At a stationary observation time, for a fixed tolerance d_f>0:

```
Pr(abs(e_f) <= d_f)
  = F_N[(d_f-mu)/s_f] - F_N[(-d_f-mu)/s_f]
```

F_N is the standard normal cumulative distribution function. When sigma_f=0, the stationary error is deterministic and should be evaluated directly. A study must select its tolerance, required probability or MSE limit, retention duration and readout independently of the result. None is numerically calibrated here.

This Ornstein-Uhlenbeck example is a local model of signed error, not a bounded structural fraction or a nonnegative distance. Its Gaussian support is unbounded. A high pointwise retention probability does not guarantee that the error remains inside tolerance throughout a long interval. Continuous retention requires a first-exit or finite-horizon analysis. Multidimensional pattern identity needs an explicit error geometry and covariance model; one coordinate cannot certify an arbitrary memory.

Changing targets, correlated disturbances, nonlinear correction and shifting cellular representations may invalidate the example. A trace readout should allow a declared form of representational change while retaining independently measured content. Neither these equations nor a good fit establish that epigenetic processes perform the modeled correction. They supply a dimensional and probabilistic replacement for the earlier shorthand, to be compared with ordinary maintenance models.

A.4 Structural coupling and phase effects

The candidate bridge adds a structural contribution to membrane or network dynamics [T4]:

```
Y(t) = integral_0^infinity h_cyto(r) M_OS(t-r) dr
theta_E(t) = theta_E0 - eta_E Y(t)
```

The kernel must carry the units converting the measured mode into `Y`; `eta_E` converts `Y` into the chosen threshold unit. These coefficients must be estimated. The notation does not turn an unspecified intracellular process into a measured causal pathway.

An optical carrier cannot itself be EEG gamma. An intensity envelope, beat or electrogenic transduction would require a physical route with adequate amplitude and timing. A sustained difference-frequency signal additionally requires suitable coherence and detection/transduction; writing two frequencies does not establish those conditions. Low-frequency structural or electrical alternatives require their own measurements and are not excluded by a failed freely propagating photon budget.

For a specified slow phase difference, a stochastic Adler model is:

```
dpsi = [Delta_omega - K sin(psi)] dt + sqrt(2D) dW_t
```

`K >= 0` and `Delta_omega` are rates, `D` is phase-diffusion strength in radians squared per second, and `W_t` is a Wiener process. In the noiseless model a stable fixed phase exists for `abs(Delta_omega)<K`. Equality is a marginal boundary. With noise, the archived rule `K > abs(Delta_omega)+D` is not a universal locking threshold. Phase slips and finite-window phase concentration depend on the parameters and observation period. Comparisons need matched common-drive and surrogate controls; increased phase-locking value alone does not identify structural coupling.

A.5 Reciprocal coupling, stability and identifiability

A local continuous-time approximation to two coupled states is:

```
de/dt = -a_e e + eta y + B_u u
dy/dt = -a_y y + zeta e + C_u u
K_loop = eta zeta
K_mag = sqrt(abs(K_loop))
```

For state units `U_e` and `U_y`, `eta` has units `U_e/(U_y s)` and `zeta` has units `U_y/(U_e s)`. Their product has units `s^-2`; `K_mag` has units `s^-1`. These units apply only to this two-first-order-state approximation. The original Rung 0 model couples into `dE/dt` and `d^2Y/dt^2`, so its product instead has units `s^-3` under physical time. Models with different orders or normalizations require separate coupling definitions; their numerical K values are not interchangeable. Report the sign of `K_loop` separately. Taking an absolute value must not erase the distinction between reinforcing and opposing feedback.

Under a constant nonzero rescaling `y_new=c y`, the coefficients become `eta/c` and `c zeta`; their product is unchanged. This explains one reason directional coefficients may not be separately recoverable without calibration. It does not guarantee that even the product is identifiable from a particular recording, sampling scheme or model.

For this two-state, instantaneous system with positive decay rates, local stability requires:

```
a_e a_y - eta zeta > 0
```

The trace is already negative. A sufficiently large positive loop destabilizes this linearization; a coupling magnitude is not automatically a measure of stable persistence. Delays, additional states and nonlinear saturation require their own analysis.

For the discrete model in the human translation [T5], stability instead requires all transition-matrix eigenvalues inside the unit circle. Discrete coefficients depend on sampling interval and should not be equated numerically with continuous-time rates. Shared inputs, correlated errors, observation noise and constructing `y` partly from `e` can generate apparent reciprocal prediction without a separate causal state. Source identity, independent observations and interventions matter as much as fitted coefficients.

A.6 What persistence would establish

A transient can decay slowly without being an attractor. Recurrence means returning near an earlier observed state; common input can produce it. An attractor is a property of specified dynamics and a basin of initial conditions, not a label earned by repeated EEG features. Persistence during continued stimulation differs from persistence after the drive ends.

Topo-OSM can operationalize human state geometry [T5], provided variables, distances, graph construction and controls are declared. Such geometry is not automatically the intracellular structure proposed by OSM. The reconstruction keeps both questions available without treating one as evidence for the other.

Independent numerical checks accompany this reconstruction for bounds, response peaks, illustrative refresh equilibria and coupling rescaling. Finite checks can expose implementation errors; the derivations establish the stated conditional results. Neither supplies fitted biological values, validates a memory substrate or demonstrates a new human effect.

A.7 Updating alignment and predictive validity

These **candidate operational definitions** refine D198 [T6]. They do not define all meaning or establish an empathy endpoint. They are mathematical specifications for prospective testing, with no change to historical Workbench scores.

A.7.1 Distributional updating

For participant i and experience u, define probability distributions q_pre and q_post over the same declared variable z. With densities relative to the same measure, use:

```
U_i(u) = D_KL(q_post || q_pre)
       = integral q_post(z) ln[q_post(z)/q_pre(z)] dz
```

Use a sum for a discrete state space. U_i is nonnegative and equals zero exactly when the distributions agree almost everywhere. Natural logarithms give nats; base-two logarithms give bits. KL is asymmetric and is not a distance metric. Finite divergence requires absolute continuity of q_post with respect to q_pre and finite integral. Positive posterior probability on a set assigned zero prior probability produces infinite divergence rather than a silently repaired finite value.

Collect prior judgments before exposure and posterior judgments afterward about the same propositions or outcomes. Specify normalization, probability elicitation, missingness and any probability floor before collection. A floor changes the reported distributions and must be disclosed. If a fitted latent-state model supplies these probabilities, freeze its likelihood, fitting procedure and calibration on separate development data. Two elicited distributions need not satisfy Bayes' rule; call their divergence reported-belief updating unless the Bayesian model is actually specified and tested. Post-only ratings cannot recover an unmeasured prior.

The equation is closely related to established Bayesian-surprise formulations [R9]. It is not a claim of mathematical priority. Large U_i can accompany inaccurate updating; small U_i can coexist with recognition and strong personal relevance. Estimation noise can also create positive divergence when true change is absent. Quantify reliability and uncertainty before interpreting a small value, and avoid treating fitted latent probabilities as direct observations of neural states.

A.7.2 Alignment in common coordinates

Use a declared map F_i to place each person's measured coordinates in a common space. Then define a change vector and metric cosine:

```
m_i = F_i(eta_i,post) - F_i(eta_i,pre)
A_ij = (m_i^T G m_j) /
       sqrt[(m_i^T G m_i)(m_j^T G m_j)]
```

G is a fixed symmetric positive-definite matrix, and both vectors must be nonzero. The Cauchy-Schwarz inequality in the G inner product gives -1 <= A_ij <= 1. The metric must account for coordinate scales and units. Report the two update magnitudes separately; equal direction does not imply equal change. Zero-vector alignment is undefined. Near-zero or poorly measured vectors require a prespecified reliability rule and uncertainty analysis, not an arbitrary denominator constant that manufactures a value.

Freeze mappings, metric and feature selection independently of the evaluation outcomes. Independently fitted participant embeddings are not automatically comparable. Under an invertible linear coordinate change m'=A m, preserve the score with G'=A^(-T) G A^(-1). Arbitrary nonlinear transformations do not preserve finite differences. This cosine is also not automatically the information geometry of the KL divergence; a common notation does not establish that connection.

The result measures update alignment. Shared input, persuasion and response style can generate it. Test an empathy interpretation against independently measured empathic accuracy and appropriate shared-input controls [R10]. Correlation with a target's reported state should be accompanied by level-error checks where relevant; matching a trajectory while consistently misestimating its level is a different result. A target's report is an imperfect criterion, and concern or helping must be evaluated separately.

A.7.3 Predictive validity on independent outcomes

D198's truth-tracking intention is retained as a narrower, testable improvement in prediction. Let p_pre,j and p_post,j be forecasts for the same externally assessed binary outcome y_j. Both forecasts must be fixed before y_j is revealed and before any scoring feedback. For n accepted, identically paired outcomes:

```
L_Brier(p,y) = (p-y)^2,                  0 <= p <= 1, y in {0,1}
Delta V_hat = (1/n) sum_j [L_Brier(p_pre,j,y_j)
                         - L_Brier(p_post,j,y_j)]
```

Positive Delta V_hat indicates lower observed prediction loss after exposure on this sample. This is an operational predictive-validity score, not a general measure of truth. For a Bernoulli outcome with true probability r, the expected Brier loss is (p-r)^2+r(1-r), uniquely minimized at p=r. That is the proper-scoring justification [R11]. A preregistered logarithmic loss is another option, with its zero-probability behavior made explicit. Do not select a score after seeing which one favors the hypothesis.

Pre and post forecasts must face the same outcome distribution and the same scoring key. If latent beliefs q are translated into forecasts p, declare and freeze that predictive mapping too. Test items, later facts or other outcomes used for evaluation must be unavailable to fitting, feature selection and participant feedback at the time of prediction. Probability change by itself says nothing about whether the error improves. For example, moving from p=0.5 to p=0.9 reduces Brier loss by 0.24 if y=1, but increases it by 0.56 if y=0. These are arithmetic examples, not experimental observations.

Report paired uncertainty at the participant and item level appropriate to the design, missing and excluded outcomes, calibration and any measurement error in the outcome. Overlapping windows and repeatedly scored answers are not new independent observations. Precision too low to exclude a meaningful improvement leaves the claim unresolved. Recall of a fictional detail supports story accuracy only; a claim about real-world validity needs independently verifiable real-world outcomes.

A.8 A dimensionally consistent meaning-arc candidate

D205's original `M_arc = Var[SMD] + max|dSMD/dt|` adds a dimensionless variance to an inverse-time quantity. A repaired candidate declares a bounded dimensionless coding trace `S(t)`, a fixed observation interval `[0,T]` and a positive reference duration `T0` [T8]:

```
S_mean = (1/T) integral_0^T S(t) dt
S_dose = integral_0^T S(t) dt
V_S = (1/T) integral_0^T [S(t) - S_mean]^2 dt
D_S = T0 * max_t |dS(t)/dt|
Arc_candidate = a V_S + b D_S
```

`S_mean` is a mean, while `S_dose` has units of time and accumulates coded exposure; they must not share a label that implies the same quantity. Both arc components are dimensionless. Freeze `T0`, nonnegative weights `a,b`, coding rules, sampling grid, smoothing and derivative estimator before evaluation. Their values are not fitted here. A discontinuous idealized trace may have no finite classical derivative; the specified discrete estimator is then a different measured quantity. Missing intervals require explicit support handling rather than invented continuity.

Report `V_S` and `D_S` separately even if a composite is used. The repair makes units consistent, not the measure scientifically valid. Stimulus coding and participant update are distinct quantities, and the composite must not be validated solely against ratings already used to create its trace.

A.9 Constructive meaning: denominator and sign policy

D203 includes signed physiological change in a cost denominator. A reduction can make that denominator approach zero or become negative unless its allowed range is constrained. The following alternative makes that policy explicit [T7]:

```
b_j = max(0, Delta c_j / s_j), with fixed s_j > 0
D_cost = epsilon + sum_j w_j b_j
CME_candidate = [Delta AVO * T_track * U_flex * C_participation] / D_cost
```

`Delta AVO` is change in Adaptive Viable Optionality, the source's proposed measure of usable, viable action options. `T_track` denotes feedback-tested predictive validity, `U_flex` feedback-sensitive strategy revision, and `C_participation` reported non-coerced engagement. None is supplied by EEG alone; each needs a declared measurement and scoring rule. This does not repair all possible definitions of AVO itself.

Use dimensionless burdens, fixed nonnegative weights and a declared `epsilon > 0`. `Delta c_j` is a signed cost change and `s_j` a positive scale in the same units. Already nonnegative error, randomness or pressure measures can enter through separately specified normalizations. This changes the original denominator and must receive a distinct definition/version. Neither `epsilon` nor the weights are estimated in this document. Gamma_phys in the source denotes proposed regulatory load, not gamma-band EEG power, joules or directly measured energetic expenditure.

Preserve signed physiological changes and their units in the report. Reduced burden is not deleted from the evidence; it simply cannot make the denominator singular or reverse the score's sign. `Delta AVO` may remain signed. Declare bounded `[0,1]` definitions for the three multiplicative factors, and leave the composite unavailable when a required input cannot be estimated. A participation rating does not replace actual consent. The signed predictive improvement Delta V_hat in Appendix A.7 is not the bounded T_track factor. Keep Delta V_hat separately reported; substituting it into this composite would require an explicitly versioned, prospectively specified mapping and a new validation.

Holding the numerator fixed, increasing the positive cost denominator reduces the score's magnitude; it is not a monotonically decreasing penalty for negative outcomes. Report benefit and burden separately rather than interpreting a less-negative score caused by greater burden as an improvement.

The same observations may contribute to AVO, its costs and the additional factors. Such reuse must be disclosed, with component-level results and held-out outcomes used to test incremental value. A positive denominator and dimensional consistency do not establish adaptive benefit, truth or a person's worth. A uniform bound on the whole score also requires a bound on its numerator.

Appendix B. Decoder Gate technical specification

B.1 Original quantities and unresolved definitions

The earlier organizing expression was:

```
MR(t) = D_gate(t) * sim(phi(u(t)), P(t))
```

Here `u` is stimulus input, `phi` a proposed representation, `P` learned prior structure, `sim` a similarity function and `D_gate` hypothesized availability. Units, measurement procedures and scale constraints remain insufficiently specified to estimate a biological gate. A weak response could reflect weak stimulation, measurement noise or an insensitive endpoint rather than a closed gate.

D223 additionally proposed:

```
Y(t) = G_R(t)[S(t)] + epsilon(t)
M_access(t) = I(M; Y(t))
```

The probability distributions, sampling model and mutual-information estimator require specification. Task accuracy is not numerically identical to mutual information. Multiplication by attention or an autonomic composite does not establish compatible units, a calibrated probability or a physiological energy budget. A signed standardized autonomic score is not automatically a nonnegative gate weight. Neither new simulation computes these original quantities [S2, S3]. Measurement quality and participant comprehension are separate variables.

B.2 Context-response model and identifiability

For the October 29 isolated-event model:

```
mu_j(c,s,tau) = n_j(tau) + f_j,tau(s)
             + b_j(tau)*(c - 1/2)
             + theta_j(tau)*(c - 1/2)*s
```

Surprise takes values 0, 1/3, 2/3 and 1; `n` represents matched nuisance inputs. The operational null is `theta=0`. Continuous traces sum finite causal event kernels, nuisance effects and correlated residual noise. Event powers and products precede convolution. Competitors share nuisance inputs and response bases. Surprise, surprise squared and surprise cubed span arbitrary additive values at the four sampled levels. This does not cover every possible history-dependent alternative. [S1]

A sigmoid parameterization does not resolve identification. For binary context and `g(c)=sigmoid(a+b*c)`:

```
lambda*g(c)*s = lambda*g(0)*s
             + lambda*[g(1)-g(0)]*c*s
```

These observations identify response coefficients, not `a`, `b` and `lambda` separately without additional restrictions. A causal contrast also requires suitable assignment, overlap, measurement and confounding assumptions.

B.3 Simulation settings and uncertainty

Each context-by-surprise dataset contained six training, three validation and six test runs, each 240 seconds with 16 events, sampled at 2 Hz. Two delayed response bases had 18-second support; stationary residual SD was 0.30 with AR(1) coefficient 0.70. Cardiac and respiratory outcomes were normalized synthetic features, not raw physiology or HRV. The study represented one fixed response profile, not a human population. The first execution and corrected context-randomization control were retained; the other twelve scenarios were numerically unchanged. Twenty-three implementation checks passed. [S1]

Each behavioral dataset used 24 training, eight validation and 16 test participants, two 16-trial sessions per participant and 96 synthetic story IDs. The simulation included four expectancy levels, producing 16 cells. These are assumed labels, not measured entropy. Participant and story splits prevented their reuse across fitting and evaluation. Scaling and penalty selection were separated from test outcomes.

The positive generator's shared state had weight 1 in coherent/narrative trials and 0.25 elsewhere. The predictor did not identify a unique biological state. The behavioral interval used 500 crossed participant/item bootstrap resamples with fitted models held fixed; its uncertainty is conditional and approximate [R3]. The reliability-informed comparison used known Gaussian reliability, not a full logistic-normal oracle likelihood. Twenty-nine implementation checks passed. Sample counts are simulation settings, not a justified recruitment target. Complete-data results do not qualify missing-not-at-random answers, sensor failure or real EEG/ECG processing. [S3]

B.4 Pilot administration, timing and records

Complete practice and calibration before the first 16-story acquisition. Return approximately 24 hours later for its memory follow-up, targeting 22-26 hours, then complete the second acquisition. Return approximately 24 hours later for the second follow-up. Preserve actual delays, interruptions, assignments, arithmetic performance and missingness. Allow roughly 25-40 minutes per main acquisition as a planning estimate. [S4]

Each trial schedules a 5-second instruction, 8-second quiet baseline, four 6-second story segments, 4-second post-story quiet display and 20-second washout after the probes. Six 4-second digit displays span the story. The critical segment begins at story +18 seconds; the candidate physiological interval is critical +4 to +10 seconds, before questions.

These are software times. Browser/server timestamps do not validate physical luminance onset or submillisecond synchronization, and the pilot adds no optical flash. Physiological uses need independently qualified timing and estimator-specific support; the short response interval does not qualify short-window RMSSD or RSA. Elapsed washout does not establish recovery.

The managed pilot has its own protocol identity, distinct from the older Signal Structure x Processing Stance entry. Fingerprinted materials/runtime and separate visit acquisition identities preserve the parent study without an overnight lock. Interruption/resume, manual scores and study identity survive verified bundle reopening. Written answers and scoring keys require privacy/disclosure review before sharing.

Internal qualification traversed both acquisitions, 64 delayed answers, interruption/retry and bundle reopen/reanalysis using synthetic records. It did not validate human feasibility or question quality. Explanatory AI summarizes derived evidence; it does not grade answers or assign a biological gate probability. [S5]

Appendix C. Optical model and reproducibility testbeds

C.1 Reduced dynamics and photon transport

The October 24 model uses dimensionless excitatory and inhibitory activity envelopes `E` and `I`, metabolic state `z`, active receiver fraction `y` and downstream state `s` [O6]. `E` is not an electric-field measurement. Photon-path transmission `g` and matched rescue `r` are expressed relative to baseline:

```
h = 0.2 + 1.6 E + z
Phi_base = Phi_reference h
Phi_received = Phi_base (g + r)
k_opt = q sigma Phi_received

dy/dt = k_on (1 - y) - d y
ds/dt = (y - s) / tau_s
dz/dt = (E - z) / tau_z

tau_E dE/dt = -E + sigmoid(2.2 E - 2 I + 0.1 + u + G s + noise)
tau_I dI/dt = -I + sigmoid(2 E - 0.8 I - 0.5)
```

The assumed settings are `tau_E=0.4 s`, `tau_I=0.6 s`, `tau_s=2 s`, `tau_z=8 s` and downstream gain `G=20`. The optical mechanism sets `k_on=k_opt`. The metabolic alternative uses activity-dependent activation independent of `g` and `r`; the synaptic alternative uses a different activity dependence. Mixed and intervention-confounded mechanisms are also tested.

Ordinary deactivation is `d=1/tau_receiver`. A reverse-conversion stress case uses:

```
d = 1/tau_receiver + rho*k_opt
y_equilibrium = k_opt / [(1 + rho)*k_opt + 1/tau_receiver]
maximum_occupancy = 1/(1 + rho)
```

At the stipulated `rho=200`, maximum occupancy is approximately 0.4975%, below the hypothetical 1% target. These are assumed stress kinetics. Receiver transitions use exact two-state probabilities for rates held constant within each step; integration of the coupled activity and metabolic states remains approximate.

For an idealized isotropic source at distance `a`, with transport fraction `T_opt`, the source demand is:

```
J_source = 4 pi a^2 Phi / T_opt
```

At `a=5 micrometres`, `T_opt=1` and the reference fluence, this requires approximately `1.59 x 10^7` activation-band photons/s before multiplication by `h`. Favorable transport does not derive that emission from chemistry. Source spectrum, radiative yield, receptor-effective fraction and nonradiative losses remain biological unknowns.

C.2 Physical-feasibility simulation results

Seven mechanisms, seven conditions and 100 noise realizations produced 4,900 main trajectories, each 160 seconds long. The endpoint was mean `E` at 100-120 seconds during stimulation. In the optical world, 99% path suppression changed it by -0.07024 arbitrary units versus sham; the 95% Monte Carlo interval was -0.07072 to -0.06975. Matched rescue returned the mean close to sham, with half rescue intermediate.

One fitted activation-rate/gain pair was approximately `0.000635/s` and `32.04`, versus planted values `0.001010/s` and `20`. Prediction did not recover the individual parameters. Twenty independent numerical checks passed. The intervals and checks concern simulation noise and implementation, not biological uncertainty or parameter validation [O6].

C.3 Modeling history and what each test establishes

The Rung 0 studies examine whether a proposed reciprocal coupling can be recovered and distinguished from alternatives under specified synthetic conditions. They do not all simulate the same system. The recovered archive supplies the original simulation and stages C-I, correcting the earlier coverage statement that their executable packages had not been located. Separately named A/B packages were not identified. The account below combines static inspection of the recovered source with its saved reports; this reconstruction did not rerun the simulations [T10].

From dynamical trajectories to response features

The original simulation couples a gamma-frequency E/I oscillator to a damped Y/V oscillator. Its equations are:

```
dE/dt = mu E - omega_g I - rho (E^2 + I^2) E + eta_E Y
dI/dt = omega_g E + mu I - rho (E^2 + I^2) I
dY/dt = V
dV/dt = -2 gamma_Y V - Omega_Y^2 Y + zeta_E E + u_Y(t)
```

Defaults set the E/I frequency to 40 Hz and the Y frequency to 7 Hz. The program integrates these equations, adds observation noise and fits `eta_E` and `zeta_E` to E alone or E plus Y. This is trajectory recovery, rather than generation of a final score. Its favorable conditions include the correct model family, known drive and fixed remaining parameters; even the optimizer's initial guess is scaled from the generating coupling values. It therefore tests conditional recoverability, not blind identification of a biological substrate.

Later stages change both the equations and the observations:

Stage	Representation and question tested
C	Discrete damped E/I and Y/V rotations replace the original nonlinear oscillator. State-space filters estimate hidden Y from E. Stress tests vary noise, initialization, alternative dynamics and held-out perturbations.
D	A leaky E process couples to a second-order Y recurrence. Added noisy Y observations and perturbation designs test whether a second measurement helps distinguish coupling directions.
E	Dynamic trajectories acquire imperfect proxy gain, delay, nonlinearity and timing. Adding a proxy does not reliably recover both parameters in the saved tests.
F	An E/Y/V differential model adds synthetic calibration measurements and perturbation selection. Calibration is generated around known nuisance values; the saved report does not demonstrate reliable directional recovery.
G	An E/Y/V differential model adds velocity and energy observations and tests composite coupling and directional ratios. Improved point estimates coexist with broad parameter profiles and competing-model problems.
H	Leaky E and second-order Y trajectories, colored noise and directional perturbations prioritize composite coupling. The E-only comparison uses a deliberately weak delayed-E proxy, not a full latent-state filter.
I	The generator explicitly switches to response features. Nine algebraic summaries depend on coupling, resonance and condition descriptors; there is no time-stepped trajectory in this generator.

These changes are substantive. C-H retain dynamical observations, but they are not successive tests of one unchanged physical model. Their proxy assumptions and unfavorable results must accompany positive recovery claims. I tests identifiability in a reduced feature representation, including programmed differences between the full model and alternatives [T10].

Later discrimination and reproducibility tests

Rung 0J creates six perturbation conditions with three algebraic response summaries each, adds noise and fits competing models. It does not generate raw neural trajectories. Its positive-coupling specification is:

```
K = sqrt(eta_E zeta_E)
K_loop = eta_E zeta_E
```

Recovering K does not separately identify `eta_E` and `zeta_E`. Across 60 synthetic trials, recorded K estimates met the 10% and 15% error criteria; the full model won 314/360 held-out perturbation folds. Rung 0K packaged an internal clean-room replay, not an independently authored external replication [L1].

Stage	Test and recorded result
0L	No-loop alternatives exposed overly permissive full-model preference.
0M	Stronger alternatives and a combined predictive/significance rule gave zero final false locks in tested nulls, with positive sensitivity 0.600.
0N	Repaired null calibration gave an approximately 0.967 positive lock rate and zero final locks in tested nulls.
0O	Locked choices with fresh seeds gave 57/60 positive locks and 0/350 null final locks.

N/O directly sample coupling estimates and model-loss scores from declared distributions, including generator-specific advantages. Null thresholds use the known generator category. Consequently, 0O is a stipulated-generator benchmark, not 95% sensitivity and zero false positives for an unknown biological recording. A deployable rule must operate without that privileged identity and be evaluated on the observations it actually receives [L1].

The October 24 physical-feasibility simulation introduces a separate photon budget and receiver-state model. It neither supplies the missing biological identity of early Y nor retrospectively validates the response-feature and score tests. Conversely, those later abstractions do not erase the early trajectory simulations. Preserve each model's equations, observations, parameter assumptions, seed policy and selection rules. Recovering a coupling, predicting held-out synthetic responses and identifying a biological mechanism are different achievements [T10, O6].

Appendix D. Earlier reconstruction comparisons and terminology

D.1 Reconstruction scopes retained separately

The separate September 29 eyes-open/eyes-closed Zuna development Run 2 evaluation used eight blocks and approximately 400 nominal scored seconds per channel after its final comparison filter. Mean NMSE was 1.0871 for Zuna, 4.1902 for spline and about 1.0000 for fixed zero. The 74.1% reduction relative to spline was a prediction-error comparison, not a percentage of artifact removed. Zuna did not beat fixed zero overall [Z1].

The September 26 post hoc bench comparison on Run 1 favored interpolation: NMSE 0.6588 versus 1.2118 for Zuna. A separately executed boundary-corrected preparation gave 0.6751 versus 1.1757. Both retained the original 102,456-sample scoring mask, but prepared inputs and comparison targets changed. This is not an isolated model improvement. These results must not be pooled with the later sparse four-recording comparison or used to restore an unavailable original endpoint [Z1].

D.2 Terms used in this document

Term	Meaning in PRAYCG
Acquisition eligibility / analysis availability	Permission and support to record are distinct from the actual inputs required for an endpoint.
ALS/PT19	Physical light-sensor timing channel.
CAI / SID	Controlled Access-Integration Index / State Integration Density; versioned derived-feature measures.
CET / EET	Cinematic Entrainment Tracking / Endogenous Echo Tracking in the declared exploratory implementation.
DGA	Specify whether the term refers to the receiver-state hypothesis, historical composite, simulation or behavioral pilot.
Gamma Scalpel	Artifact-forensics and matched gamma comparison, distinct from signal cleaning.
HED	Structured event descriptions; syntax validation is separate from event and timing verification.
MRED	Meaning Recognition / Encoding Dissociation; operational recognition-like and later integration-like outcomes.
NAST / NAS	Narrative-absorption candidate/component, not direct default-mode-network measurement.
NIP	Narrative Immersion Proxy; an exploratory measure.
OptoPING	Proposed electrical/optical/persistent-state feedback family.
PRV / HRV	Variability derived from qualified pulse or cardiac observations respectively.
Reanalysis / replication	New processing of an existing acquisition / a new acquisition testing a stated result.
TSP	Temporal Semantic Proxy; proposed regional/feature contrast, not a localized STG-to-MTG exchange measurement.
Zuna output	Reconstructed derivative rather than an observed electrode measurement.
OSM-CELL	The candidate intracellular encoding/storage/retrieval mechanism; not identified by EEG alone
OSM-TOP / Topo-OSM	Human network-state organization, persistence and reactivation; geometry must be specified
Structural refresh	A proposed maintenance process under turnover; not proof of pattern fidelity
Update alignment	Directional similarity in a declared common state space; not a validated empathy measure

Appendix E. Operational measurement definitions

This appendix restores the original master's measurement detail while retaining the later distinction between a candidate construct, its numerical implementation and observed evidence. CAI/SID concerns coordinated access-like and integration-like features; CAA describes autonomic trajectories; Micro Handoff asks whether one activation proxy rises before another. Their names preserve the research questions. None makes successful computation evidence of meaning, memory formation or a cellular mechanism. The formulas below document existing versioned definitions; they introduce no new score or fitted coefficient [M1-M3].

E.1 CAI/SID v0.2: components and normalization

For a feature with usable Baseline 1 scale, the original robust transform is:

```
z_B1(x_t) = 0.67448975 * [x_t - median_B1(x)] / MAD_B1(x)
```

The frozen worker clips standardized values to [-6,6]. Its legacy configuration permits a recorded full-run fallback when baseline support is insufficient, and can use a standard-deviation scale when MAD is unusable. The managed diagnostic passes its selected configuration to that core; management alone does not disable fallback. Paired Signal Review instead fits upstream features on original matched data, derives the CAI reference from original Baseline 1, applies that reference to the candidate and disables full-run baseline fallback. Micro Handoff has its own no-full-run-fallback rule. A zero or unavailable scale cannot justify silently supplying a neutral value. These are within-run deviations, not population norms.

Write WM(z;w)=sum(w_j*z_j)/sum(w_j) over explicitly available finite inputs, retaining their declared positive weights. With all listed inputs available, the original fast and slow definitions are:

```
L_E = 0.45*zMeaningGamma + 0.35*zTSP + 0.20*zNIP
      - 0.10*max(zVisualGamma,0)
L_Y = 0.50*zThetaIntegration + 0.30*zNAS + 0.20*zPosteriorAlpha
E = sigmoid(L_E)
Y = sigmoid(L_Y)
```

The implementation uses WM for available feature combinations; optional NIP's absence therefore renormalizes the fast positive-input weights. Required source columns and readiness rules still apply. Missing optional visual evidence is disclosed. E and Y are unitless bounded activation proxies, not calibrated probabilities. Their feature names do not establish that gamma represents meaning or theta proves storage.

The retrospective component is:

```
Y_future(t) = mean{Y(s): t+8 <= s <= t+30,
                         s belongs to the same contiguous block}
e_plus = max(E-0.5,0)
y_plus = max(Y_future-0.5,0)
K_product = 2*sqrt(e_plus*y_plus)
K = 0.70*K_product + 0.30*max(corr_local(E,Y_future),0)
```

The frozen configuration requires at least five future observations and ten paired observations for correlation within its 31-second local window. Insufficient future support stays missing. When correlation is non-estimable, K equals K_product without the .70 multiplier. e_plus and y_plus lie in [0,.5]; K lies in [0,1]. The positive-amplitude product is zero for jointly below-midpoint values, but positive correlation can still contribute. For example, perfectly correlated values confined to .10-.40 give K_product=0 and K=.30. This arithmetic illustrates implementation behavior, not semantic integration.

For the support and penalty terms, define a different rescaling:

```
excess(v) = clip(2*(v-0.5),0,1)
A = sigmoid(WM([zArtifact,zHighFrequency,zPeakToPeak];[.55,.25,.20]))
X = sigmoid(zTask)
Support = mean(excess(E),excess(Y),K)
Penalty = mean(excess(A),excess(X))
CAI_core = Support*(1-Penalty)
```

Direct slow support uses current Y; delayed Y_future enters the loop. This distinction is part of v0.2. All support and penalty inputs lie in [0,1], so CAI_core lies in [0,1]. With current E and Y below .5, K=.30 and zero penalty, CAI_core can equal .10. Hard-invalid windows and unavailable required components remain missing. Equal support/penalty weights are design choices, not fitted biological constants.

DGA/contextual reliability and CAA remain separate sensitivity or convergence outputs. They must not be silently multiplied into primary CAI. Derived NAS can reuse CAI inputs; its provenance and dependence must remain visible.

E.2 SID measures supported time

The original master's simple valid-row mean and full-interval integral agree only with appropriate equal sampling and complete support. The later managed layer uses nonoverlapping accepted duration:

```
T_valid = sum(dt_i over valid observations)
CAI_load = sum(CAI_i*dt_i over valid observations)
SID = CAI_load/T_valid
occupancy(tau) = sum(dt_i where CAI_i>=tau)/T_valid
```

For 100 seconds of available observation support, with 20 valid seconds at CAI=.8 and the remaining 80 seconds invalid for CAI, load is 16 score-seconds and the valid-time ratio is .8. Filling those 80 seconds with zero and dividing by 100 yields .16; that treats missingness as low evidence. This example illustrates the denominator only; its 20% coverage would fail the managed 70% support rule, so a qualifying SID would not be reported. Report coverage separately. Zero valid duration makes SID and occupancy unavailable. Duplicate timestamps, overlapping windows and long gaps require explicit support accounting. The managed 70% valid-support threshold governs computational estimability; it does not validate the construct. The added valid-time layer does not inherit an uncertainty interval merely because the legacy core reports one.

E.3 CAA: intervals, adjacent pairs and respiration

For observed beat intervals in milliseconds:

```
IBI_i = RR_i/1000                [seconds]
HR_i = 60/IBI_i                 [beats/minute]
RMSSD_W = sqrt(mean((IBI_(i+1)-IBI_i)^2 over accepted pairs in W))
pNN50_W = fraction of accepted pairs with abs(IBI_(i+1)-IBI_i)>.05 s
```

Pairs must be adjacent in the original beat ledger. Removing an invalid interval and then differencing the remaining array can join unrelated neighbors. Gaps and connection boundaries remain explicit. SDNN requires declared valid normal-to-normal intervals and a specified standard-deviation estimator. Raw interval variability is not automatically qualified NN-HRV; reciprocating sampled HR does not reconstruct observed beat events.

The older 30-, 60- and 120-second window choices remain historical analysis settings, not guarantees of scene-level resolution. Respiration phase, rate, depth and events support interpretation; a breath event can be physiology and an artifact risk. The original RSA correlation/coherence shorthand is not numerically interchangeable with current harmonic regression against measured phase or optional NeuroKit RSA. Source

pairing, timing and estimator-specific support are required. PPG-derived PRV remains distinct from ECG-derived HRV.

E.4 Micro Handoff: temporal meaning

Micro Handoff v0.1 uses trailing 250 ms temporal-gamma estimates and trailing 500 ms theta estimates, followed by declared trailing smooths. Its 25 ms output grid is a refresh schedule. Overlapping rows do not supply independent spectral observations or 25 ms biological transfer resolution.

The fixed 100-1000 ms lag bank separates state agreement, positive directional change and activation-weighted density. Low-low can agree; stationary high-high can contribute density without a directional rise. The primary candidate averages the declared lag bank rather than selecting a favorable condition-specific lag. Because the slow value is evaluated at t+tau, directional results are offline. Feature windows and smoothing contribute to fitted lags; these scalp groups do not directly measure an STG-to-MTG transfer. Synthetic lag recovery qualifies stipulated software behavior, while mixed historical null results retain their original limitation. Micro Handoff remains exploratory and separate from the primary conclusion chain [M3].

Appendix F. Disposition of the original master

All 36 numbered sections of public v1.1 are accounted for below. Retained means that the research purpose remains; it does not mean every historical sentence, calculation or operational instruction was copied unchanged. The original edition remains the source for its historical wording. Chapters 2 and 8-9 and Appendices A-C retain the subsequent theory and mathematical developments.

Original	Topic	Location in v3	Treatment
1	Executive summary	Ch. 1	Research question and reproducible pipeline retained; software scope updated to distinguish the original protocols from the broader current platform.
2	Why PRAYCG exists	Ch. 1; 3-4	Sensory, local-structural, narrative, task, after-state and access distinctions restored as competing research questions.
3	Public naming and acronym boundary	Ch. 1; App. D	Empirical scope and project terminology retained without claims derived from the project name.
4	v0.95 development map	Ch. 10; App. F	Detailed historical map retained in [P0]; current software described separately under [W5].
5	Theory in one page: meaning as gated interaction	Ch. 1-2; 8	Access-by-prior-state hypothesis retained in the receiver account; gate quantities remain operational proposals rather than calibrated probabilities.
6	Availability under load	Ch. 1-2	Coherence, update capacity, persistence and load retained as organizing concepts; the uncalibrated quotient is omitted.
7	Protocol families in v0.95	Ch. 3-4	PRAYCG3, PRAYCG4 and SMG rationales retained; historical design and current operational requirements distinguished.

Original	Topic	Location in v3	Treatment
8	Branch doctrines	Ch. 3-4	Each branch's intended preservation and disruption restored. Override remains compatible with meaning, and controls do not establish its absence.
9	Counterbalancing and Williams-sequence variants	Ch. 4	Assignment logic and period/carryover limitations retained; available sequence templates distinguished from an achieved balanced sample.
10	Media procurement and stimulus selection	Ch. 4; 10	Lawful sourcing, stimulus constraints and familiarity retained; obsolete operational details remain in [P0].
11	MediaPrep and stimulus generation	Ch. 4	Visual/audio review and content identity retained; planned transformations distinguished from verified final media.
12	Anchor doctrine	Ch. 4; 6	Frame verification, prospective locking, hashes and source identity restored. Hindsight annotations remain exploratory.
13	Hardware and acquisition architecture	Ch. 5	Modality-specific requirements and physical timing rationale retained; device descriptions updated to the scoped current evidence.
14	Protocol runner and Control Center	Ch. 10	Manifest/lock rationale and continuity across the study retained under current Workbench terminology.
15	Self-report and neurophenomenology	Ch. 3-4; 6	Participant experience retained as a separate observation; dependence is explicit when a rating also enters a composite.
16	Master Comprehensive Analysis Suite v0.95 posture	Ch. 6	Original question chain restored. Historical tiers and later registry differences are distinguished from prospective endpoint validation.
17	Core analysis modules retained from v1.0	Ch. 6; App. E	Individual purposes restored; legacy GammaScalpel composites distinguished from current measured-reference Gamma Scalpel.
18	Expanded confound defenses	Ch. 6; 8	Confound questions and limits restored; DGA theory and results developed separately from the legacy score.
19	CAI/SID v0.2	Ch. 6; App. E	Purpose and components retained; valid-time normalization, positive-excess scaling and current/future terms clarified without changing the worker.
20	CAA v0.1 / Continuous Autonomic Arrays	Ch. 6; App. E	Time-resolved purpose retained; interval units, adjacency, window support and distinct respiratory-coupling estimators clarified.
21	Micro Handoff v0.1	Ch. 6-7; App. E	Three quantities, fixed lag policy and mixed historical status retained; software evidence restricted to the cases actually tested.

Original	Topic	Location in v3	Treatment
22	Visualizer and offline interpreter	Ch. 6; 10	Evidence inspection and missingness retained; current views and reports preserve source identity without upgrading the underlying claim.
23	Baseline 2: promise and danger	Ch. 3; 6	Cumulative after-state restored explicitly, with Target carryover, task relief, fatigue and respiration as competing attributions.
24	Pilot-derived archetypes	Ch. 2-3; 6-7	Recognition-dominant, peak-plus-density, resolution and access dissociations retained as candidate patterns rather than validated classes.
25	Topo-OSM, memory, and mechanism quarantine	Ch. 2; 9; App. A	Memory and maintenance hypotheses reconciled with the later theory; measurement scales and mechanism boundaries remain explicit.
26	Rung 0 / synthetic identifiability	Ch. 8-9; App. B-C	Synthetic-test rationale retained and later model-specific cases documented; scoped recovery distinguished from construct validity.
27	Confirmation-grade decision rule	Ch. 11	Quality, anchor and replication requirements retained as claim-specific criteria. Checklist completion and universal Target dominance do not establish confirmation.
28	Prospective study roadmap	Ch. 11	Reproducibility, individual replication, feasibility and prospective group evaluation retained, with independent outcomes.
29	File provenance and folder map	Ch. 10; App. F	Record identities and relationships retained; exact historical folders remain attached to their original releases.
30	Template files	App. F	Template roles summarized for reports, anchors, features and derived outputs; exact schemas and compatibility remain with the selected release.
31	Safety, ethics, and public-release boundary	Ch. 1; 10	Nonclinical scope, consent, data protection, stimulus rights and transparent limitations retained.
32	Current discovery map: hypotheses, not proofs	Ch. 2-3; 6	Dissociations and access claims retained as hypotheses; conceptual plausibility distinguished from observed findings.
33	What PRAYCG may eventually become	Ch. 1; 11	Open-workbench purpose and competing-explanation questions retained without promises of future validation.
34	Recommended next study design	Ch. 4; 11	Preparation, measurement qualification, frozen endpoints and separate exploratory outputs retained with current hardware evidence.

Original	Topic	Location in v3	Treatment
35	Minimal glossary	Ch. 6; App. D	Terminology retained across the methods account and glossary; legacy DGA, MeaningGamma and TSP distinguished from current measurement procedures.
36	Closing synthesis	Ch. 1; 11-12	Availability, recognition, integration and persistence retained as the organizing chain, with conclusions bounded by the existing evidence.

Historical software and record templates

The v0.95 development map explains the origin of the original study pipeline. It is retained in [P0], rather than presented as the current release inventory. V3 describes Public Alpha 11.11.0 under [W5]; a software version and a theory edition identify different things. Historical output folders and schema names remain tied to the release that produced them.

The original templates served the following research roles. A reproduction must use the actual versioned schema and field definitions from its selected worker, rather than infer them from this summary.

Record family	Research information to preserve
Familiarity and experience reports	Participant/session identity, prompt version, scale, response, missingness and collection time
Scene map and anchor lock	Source stimulus hash, frame/content time, anchor rationale, prospective lock or later annotation status
Events and acquisition evidence	Actual event timing, stream/channel identity, units, gaps and physical synchronization evidence
Feature and artifact tables	Worker/settings version, represented interval, calibration, available inputs, exclusions and quality support
CAI/SID and CAA outputs	Components, valid duration, window support, physiological units, missingness and method-specific eligibility
Micro Handoff and KHT outputs	Feature definitions, lag/window policy, normalization, candidate selection and separate outputs
Analysis reports and bundles	Input hashes, plan/amendments, code/environment identities, execution states, evidence links and privacy scope

These roles are preserved without assigning invented fields to old files. Display-name corrections or new templates do not change an old run identity or upgrade its evidential status. Exact compatibility and migration rules belong with the selected release.

Appendix G. References and source guide

Project records identify the implementations, assumptions and saved outputs summarized here. Literature references identify relevant prior work and measurements. Full filenames, available hashes and the prior-to-current section map are retained in the accompanying **Source and Revision Ledger**. Local records are not represented as public downloads. This edition adds source comparison and implementation clarification. Newly summarized software and retrospective equipment evidence is identified as [W5] and [H3]; no participant experiment or archived scientific analysis was rerun for v3.

G.1 Project records

[P0] Banks, H. *PRAYCG Master Theory and Protocol*, public v1.1 aligned with v0.95. Original 35-page research account, retained unchanged. Source of the restored experimental explanation and section-disposition map.

[S0] Banks, H. *PRAYCG Master Theory and Protocol*, reconstructed public v2.3 (November 1, 2026) and reconciled public v2.4 (November 7, 2026). V3 preserves their theory, corrected mathematical specification and scoped evidence while restoring the detailed narrative research account. Source identities appear in the revision ledger.

[S1] Decoder Gate Falsification Simulation, October 29, 2026. Package `DecoderGate_Falsification_Simulation_20261029`: assumptions, simulation, corrected results and verification. Thirteen scenarios; first execution retained separately.

[S2] Decoder Gate original-source review, October 30, 2026. D223 *Opto Gate / Decoder Gate Theory*, D224 *Decoder Gate Experimental Protocol* and D225 integration register; source ledger and review in the behavioral-validation package.

[S3] Decoder Gate Behavior Validation, October 30, 2026. Package `DecoderGate_Behavior_Validation_20261030`: prospective protocol, assumptions, simulation, results and verification for eleven scenarios.

[S4] Decoder Gate Pilot Pack v1, October 30, 2026. Version notes, settings, participant bank, counterbalance lists and scoring keys in `DecoderGate_Pilot_Pack_v1_20261030`.

[S5] Public Workbench Alpha 11.10.0, Decoder Gate integration. `Decoder_Gate_Pilot_v1_0`, packaged-file ledger and synthetic traversal/reopening records.

Records [W1-W4] identify the Public Alpha 11.10.0 snapshot. [W5] and [H3] identify the newly reviewed 11.11.0 release and equipment evidence. Retained components keep their own method versions; [M4] identifies the legacy methods inspected in the current release.

[W1] Release and workflow records. README, changelog, build report, installation guide and Cerelog/Decoder Gate workflow.

[W2] Protocol, stimulus and evidence contracts. `PROTOCOL_EVIDENCE_11_7.md`, fingerprint analysis contract and local DATA catalog/environment specification.

[W3] Analysis and processing workflow. Analysis orchestration registry, guided workspace, Signal Review guide, preservation/workflow guides and live/replay optical specification.

[W4] Explanations and AI authoring. Installation/workflow guide, explanations and AI-return specification, and AI-authoring workflow review.

[W5] Public Workbench Alpha 11.11.0, November 8, 2026. Current README, installation/workflow guide, 18-item scope, build verification, analysis registry, Scientific Study Review service and current protocol/runner sources. Archive SHA-256 `260342f386bc1a0d2efe0dfe2eacd4ab66fabfdbf5de0cc552e5a8dc6c1a174a`. This is a software release record, not participant validation.

[H3] Saved joint equipment recordings and retrospective detector review, November 2026. Six-minute dual-ALS, 45-second joint and sensor-swap source recordings, reviewed in the Public 11.11.0 build. The retained `HARDWARE_IMPLEMENTATION_QA.json` and timing-review output identify source hashes, threshold sensitivity and

held-out pairs. Original recordings and earlier reports remain unchanged. No new physical run was made for the release or this document.

[G1] Gamma Scalpel 2.0. Worker, measured-reference implementation, README and reference-profile guide; current compact hardware specified in the release's Cerelog workflow.

[G2] Signal cleaning and SPAR comparisons. Signal Review method records, preservation guide, synthetic preservation benchmarks, reference/control review and upstream notices.

[H1] Compact Cerelog acquisition. Current profile, managed connector, compact workflow, retained-capture replay and dual-ALS comparison records.

[H2] Hardware and timing contracts. Athena acquisition evidence, Cyton/Daisy profile, Polar and Vernier connectors, Gamma Scalpel reference guide and verified optical-profile guide.

[M1] CAI/SID v0.2. Worker, managed diagnostic specification and paired-processing adapter.

[M2] Autonomic methods. Continuous Autonomic Arrays, Signal Review physiology and shared RR-evidence implementation.

[M3] Micro Handoff and KHT. Worker/specification, synthetic and historical audits, and Master Suite event/carryover calculations.

[M4] Retained Master Suite methods and implementation. Current `MasterComprehensiveSuite_v1_6_1_CURRENT` package in [W5], including the v1.6.0 master and NIP/CET/EET workers; A-MRED v1.5.3, MRED Peak/Resolution v1.5.5 and CandidateLocal KHT v1.4 method records; v1.5.7 module-tier registry. These retain historical operational definitions. Source review does not establish scientific validation or equivalent behavior across every legacy path.

[D1] Contact, September 2026. Saved effects summary, report and subsequent readable evidence corrections.

[D2] Arrival, Snack Attack and About Time Run 2, September 2026. Three-recording review with per-recording effects, results and reconstruction summaries.

[D3] Field of Dreams and Her. Preserved public Field full-report package, Her anchor note and runner presentation-correction notes.

[D4] Historical recording inventory. August 29 full-data audit, Eternal Sunshine candidate endpoint table, interrupted About Time Run 1 and later source-linkage inventory.

[Z1] Zuna comparisons, September 2026. Contact and three-recording reconstruction summaries; earlier-scope clarification report; September 26 original and boundary-corrected interpretations.

[O6] OptoPING Physical Feasibility Simulation, October 24, 2026. Assumptions, simulation, results, audit notes and pre-run source hashes in the named package.

[L1] OptoPING Rung 0 collection. Preserved public `research/studies/optoping-rung0/` packages and `research/optoping/README.md`, including executable J/O sources.

G.2 Reconstructed theory sources

These are historical project proposals and model specifications, not independent empirical confirmation. Document IDs distinguish sources across repackaged archives; repeated identical copies are not independent revisions. Neutral archive labels and exact hashes appear in the Source and Revision Ledger.

[T1] Original OSM and amended D03. *Opto-Structural Memory Theory*; original two-page account and amended seven-page D03 account in the recovered OSF archive. Encoding-related tuning, cue-sensitive response and distributed storage. The public theory brief is retained as a separate scope statement.

[T2] OSM network-novelty patch. Ten-page theoretical framework in the recovered archive; network excitation/inhibition, calcium/plasticity gain and the distributed extension.

[T3] D196. *Structural Epigenetic Memory Refresh Theory*, three pages. Structural trace, scaffold, circuit, regulatory support and reactivation; source of the refresh architecture and the threshold statement corrected here.

[T4] D182 and Optional OptoPING Reconciliation Appendix. *Opto-Electrostructural Membrane Coupling Theory*, three pages; optional appendix, seven pages. Candidate structural-to-membrane transfer, phase effects and reciprocal modeling.

[T5] Topo-OSM reformulation. Human-scale network-state reformulation and claim-boundary notes, version 1.4.7; integration register and manuscript version 0.3. These records explicitly separate human state proxies from intracellular mechanism claims.

[T6] D198. *Meaning-Empathy State-Space Reconciliation Addendum*, three pages. State update, distributional displacement, interpersonal alignment and predictive validity. The newly supplied addendum and *Meaning Empathy State Space Reconciliation Compendium* have identical extracted text and are treated as copies of one proposal. Appendix A.7 gives the revised operational definitions.

[T7] D203-D204. *Constructive Meaning Efficiency Theory* and experimental protocol. Benefits, option generation, validity, flexibility and cost. The denominator candidate in Appendix A.9 is a new versioned formalization, not an assertion that the archived score was implemented or rescored.

[T8] D205-D207. *Semantic Meaning Gradient Theory*, experimental protocol and integration update. Stimulus distinctions, coded meaning gradients and arc measures; dimensional repair in Appendix A.8.

[T9] Working manuscript and Wild Hare synthesis. OSM working manuscript v0.5, expanded synthesis v0.2 and full formula/module report v1.1, supplied in the v0.5 update archive. Receiver/key relationship, recognition versus encoding, accommodation versus resolution and later readiness.

[T10] Recovered OSF Rung 0 archive. `cr654-osfstorage-archive.zip`: original simulation, nested C-I code and saved reports, optional reconciliation appendix and public experimental protocol. Static source review establishes model definitions; this revision did not rerun these programs or certify source-to-output provenance.

[T11] D274. *Hidden Y Taxonomy and Biophotonic Candidate Boundary*, three pages. Distinguishes mathematical, synthetic, human-transition, cellular-proxy and memory-relevant structural states.

[T12] Structural Epigenetic Memory Refresh public boundary note. *A self-refreshing attractor account of long-term memory persistence under molecular turnover*, two pages. Supplied public summary of the five-variable architecture, refresh ratio and fidelity shorthand. Section 2.3 and Appendix A.3 retain its intent while restricting the ratio to explicit dynamics and replacing the ambiguous stochastic expression.

G.3 Primary literature

[R1] Yeshurun, Y., Swanson, S., Simony, E., Chen, J., Lazaridi, C., Honey, C. J., and Hasson, U. (2017). Same Story, Different Story: The Neural Representation of Interpretive Frameworks. *Psychological Science*, 28(3), 307-319. doi:10.1177/0956797616682029. Open manuscript.

[R2] Craik, F. I. M., Govoni, R., Naveh-Benjamin, M., and Anderson, N. D. (1996). The effects of divided attention on encoding and retrieval processes in human memory. *Journal of Experimental Psychology: General*, 125(2), 159-180. doi:10.1037/0096-3445.125.2.159. PubMed record.

[R3] Owen, A. B. (2007). The pigeonhole bootstrap. *The Annals of Applied Statistics*, 1(2), 386-411. doi:10.1214/07-AOAS122. Author manuscript.

[R4] Whitham, E. M., et al. (2007). Scalp electrical recording during paralysis: quantitative evidence that EEG frequencies above 20 Hz are contaminated by EMG. *Clinical Neurophysiology*, 118(8), 1877-1888. doi:10.1016/j.clinph.2007.04.027.

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[R6] Hipp, J. F., and Siegel, M. (2013). Dissociating neuronal gamma-band activity from cranial and ocular muscle activity in EEG. *Frontiers in Human Neuroscience*, 7, 338. Open article.

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[O2] Kobayashi M and colleagues. (1999). In vivo imaging of spontaneous ultraweak photon emission from a rat's brain correlated with cerebral energy metabolism and oxidative stress. *Neuroscience Research*, 34(2), 103-113. DOI:10.1016/S0168-0102(99)00040-1.

[O3] Kojima D, Mori S, Torii M, Wada A, Morishita R, Fukada Y. (2011). UV-Sensitive Photoreceptor Protein OPN5 in Humans and Mice. *PLoS ONE*, 6(10), e26388. DOI:10.1371/journal.pone.0026388.

[O4] Zhang KX and colleagues. (2020). Violet-light suppression of thermogenesis by opsin 5 hypothalamic neurons. *Nature*, 585, 420-425. DOI:10.1038/s41586-020-2683-0. Open full text.

[O5] Tang R, Dai J. (2014). Spatiotemporal Imaging of Glutamate-Induced Biophotonic Activities and Transmission in Neural Circuits. *PLoS ONE*, 9(1), e85643. DOI:10.1371/journal.pone.0085643.

[R8] Smolen, P. (2007). A Model of Late Long-Term Potentiation Simulates Aspects of Memory Maintenance. *PLoS ONE*, 2(5), e445. Article and model. DOI:10.1371/journal.pone.0000445. A prior modeling account of reactivation and biochemical feedback; not direct evidence for the proposed OSM substrate.

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[R10] Zaki, J., Bolger, N., and Ochsner, K. (2009). Unpacking the informational bases of empathic accuracy. *Emotion*, 9(4), 478-487. Article record. DOI:10.1037/a0016551. An independent interpersonal criterion, rather than similarity of two update vectors.

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